AL-BATTĀNĪ

# Commentary on Ptolemy's *Tetrabiblos*

*Sharḥ Kitāb al-Arbaʿ maqālāt fī aḥkām ʿilm al-nujūm*

translated from the Arabic of the Escorial manuscript, árabe 969, ff. 20r–80v

This is an English translation of al-Battānī's commentary on the *Tetrabiblos*, the earliest that survives. It is made from Nadine Löhr's transcription of the Escorial manuscript in *Ptolemaeus Arabus et Latinus* (Bavarian Academy of Sciences and Humanities, 2021, CC BY 4.0). The commentary follows Ptolemy's work chapter by chapter, restating its doctrine and adding al-Battānī's own explanations, examples and procedures; the notes give the chapter of the *Tetrabiblos* in F. E. Robbins's numbering, which was used as a check on the meaning, and explain terms. Every sentence of the Arabic is translated, in the order of the text; numbers are given in figures. Words added for the sense stand in square brackets; words supplied by the editor of the transcription in angle brackets ⟨ ⟩. Words that the editor marks as additions of the copyist, such as invocations, stand in double brackets ⟦ ⟧, and words struck out by the copyist are left out. A doubtful reading of the transcription is marked (?). The folios of the manuscript are given in the margin.

The translation is in progress. This version holds the title page and chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24. It is AI-integrated work (Claude), 2026; no person has proofread it. The translation is dedicated to the public domain (CC0 1.0); the Arabic it translates is Löhr's transcription, used under its licence (CC BY 4.0).

f. 20r

The Book of the Four Treatises on the Judgements of the Stars, by the lord of the sages, Ptolemy:

the commentary of Abū ʿAbd Allāh Muḥammad ibn Jābir Abū Sinān al-Ḥarrānī, known as al-Battānī[1]

f. 20v

[[In the name of God, the Merciful, the Compassionate; God is my sufficiency.]]

## ⟨1⟩ On knowing the power of the sun, the moon, the wandering stars and the fixed stars[2]

Abū ʿAbd Allāh Muḥammad ibn Jābir al-Ḥarrānī, known as al-Battānī, said: The action of the substance of the sun is to dry and to heat, the drying being slight. As for the moon, most of the action of its power is moistening, and it changes bodies visibly, because it ripens and rots most of what is in them; it also shares a little in heat with the sun, because it receives light from it. Saturn acts mostly by cooling, and it dries a little, because of its cold and dryness. Jupiter is of a temperate mixture, for it heats and moistens together, and it produces fertilizing winds. Mars dries above all, and most of the action of its power is burning, because its nature is burning, its colour being like the colour of fire. Venus is of a temperate mixture, except that it moistens much and heats a little. As for Mercury, its moistening and its drying are equal: at some times it dries and absorbs moisture, at others it moistens, and it passes quickly from one state to the other; and it generates winds.[3]

As for the powers of the fixed stars, from their mixtures, compared with the powers and natures of the wandering stars as they were found for them when they were observed: we have set them down in the tables of the positions of the fixed stars, in a book other than this, and drawn beside each star a mark by which the wandering stars that its nature resembles are known. Some resemble in their mixture the mixture of one wandering star; some resemble the mixture of two stars equally; and some resemble one of two stars much and the other a little. Their natures resemble [the wandering stars] as the marks found there show, and for that reason we can dispense with mentioning their powers here.[4]

## ⟨2⟩ On the powers of the planets according to the configurations they have relative to the sun[5]

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The moon and the three upper planets receive increase and decrease in their particular powers according to the configurations they have with the sun. From the time when the moon appears as a crescent to the time of its first half in light, it produces moisture above all; from its first half to its full, when it is in opposition to the sun, heat; from its full to its second half in light, dryness; and from its second half to its disappearance, cold.

Saturn, Jupiter and Mars, from the time of their rising in the mornings and their appearing from under the rays of the sun to the time of their first station, moisten more; from their first station to their rising at the beginning of the night, which is when they are in opposition to the sun, they heat more; from their rising at the beginning of the night to their second station, they dry

[1] So the manuscript, «Abū Sinān», for «ibn Sinān» (PAL: sic).

[2] Ptolemy, *Tetrabiblos* I.4 and I.9. The chapters of the *Tetrabiblos* are cited in F. E. Robbins's numbering.

[3] «Fire» reads النار with the Arabic *Tetrabiblos* (PAL's T); the manuscript has العار, «shame».

[4] «Beside» reads إزاء for the transcription's إذا. Ptolemy gives the powers of the fixed stars in *Tetrabiblos* I.9. The star catalogue of al-Battānī's *Zīj*, as Nallino edits it from the Escorial codex, has no such marks.

[5] *Tetrabiblos* I.8.

more; and from their second station to their setting and disappearance, they cool more. When they mix with one another, they produce in the air very many kinds of very different conditions, and these conditions are called the causes of the stars.

### ⟨3⟩ On the powers of the four seasons of the year and of the ages of animals[6]

In spring moisture prevails, in summer heat, in autumn dryness and in winter cold. In the same way, in the first age of animals the moist substance prevails; in the second age, up to the time of the prime, heat; in the third age, when it nears the decline and begins to decrease, dryness; and in the last of the ages, when it nears dissolution, cold: like spring, summer, autumn and winter.

### ⟨4⟩ On the powers of the four angles and the four quarters of the horizon[7]

So too is the condition of the four quarters of the horizon and of the angles from which the winds blow to all sides. Dryness prevails in the quarter next to the east, and the wind that blows from it, the east wind, which is called Apeliotes, is without moisture and drying. The winds that blow from the side of the south, which are called southern winds, are hot and burning. The quarter of the west is moist, and the winds that blow from it, which are called western, soften and moisten. The quarter of the north is very cold, and the winds that blow from this quarter, which are called northern, are cold and condensing.

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The knowledge of these matters is also useful and helps to distinguish the mixtures. For it is clear that the actions of the power in the stars change with the conditions of the seasons of the year, of the ages or of the angles; that when the conditions are akin to them, their qualities are pure and unmixed, with nothing mingled in them, and their action is stronger, as the stars that heat by nature are stronger in hot conditions and the moist stars stronger in moist conditions; and that in the opposites of these conditions their power is not pure and unmixed but, being mixed with something else, weak, as the power of the heating stars is weaker in cold conditions and that of the moist stars weaker in dry conditions. The natures of the fixed stars that mix with those stars, or share with them in one of the nine configurations that they have, especially with the sun, which were mentioned earlier in this book, increase this, decrease it and change it not a little. In the same way the change in all the other conditions follows the mixing of the qualities.

### ⟨5⟩ On the fixed, the solstitial and the bicorporeal signs[8]

Of the signs, two are solstitial and two are equinoctial. The solstitial ones are Cancer and Capricorn, and the equinoctial ones Aries and Libra. These signs are given these names because when the sun comes to the point of the summer solstice it brings about summer; when it comes to the point of the winter solstice, winter; when it comes to the point of the spring equinox, spring; and when it comes to the point of the autumn equinox, autumn; and at these two points night and day are equal over the whole earth.

Of the eight remaining signs, which lie between the points of the equinoxes and of the solstices, four are fixed. They are called fixed because the season stays fixed in them in one state; they are the ones that follow the two signs of the equinoxes and the two signs of the solstices, and they are called Taurus, Scorpio, Leo and Aquarius. The bicorporeal signs are those that follow the fixed signs. They are given this name because they lie between the fixed signs and the signs of the equinox and the solstice, and share, as it were, with their beginnings in the fixed signs and with their ends in the changing signs; they are Gemini, Virgo, Sagittarius and Pisces.

[6] *Tetrabiblos* I.10.
[7] *Tetrabiblos* I.10.
[8] *Tetrabiblos* I.11.

## ⟨6⟩ On the sharing of the signs in the configurations[9]

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As for the matters of the sharings in the configurations, which are the conjunction, the opposition, the quadrature, the sextile and the trine; of the signs that look at one another and are equal in power; of the commanding and the obeying signs; and of the signs that are not joined to one another and do not share in the configurations: we have explained them earlier in this book, since [there was given] the cause that shows why these distances alone are used and none other of the geometrical things. So too the others: the contrary and associated signs are those opposite one another; the hostile signs are those in quadrature to one another; and the friendly signs are those in trine to one another, and also those in sextile, since these two configurations fall between signs of one kind, either masculine or feminine, while the first two configurations are otherwise. Moreover, the houses of the two maleficent planets are in opposition and in quadrature to the houses of the luminaries, and the houses of the two beneficent planets in trine and in sextile to the houses of the luminaries.[10]

## ⟨7⟩ On the masculine and feminine signs, their other conditions and the names given to them[11]

The signs feminine and masculine by nature we have explained in mentioning the triplicities. Some people have used another way of ordering the masculine and feminine signs, from the sign of the ascendant: they counted one masculine and one feminine, and made the beginning from the masculine; just as some people take the solstitial signs beginning from the sign of the moon, because the moon, through the speed of its motion, turns more than the other stars. Some of them divided the signs into four parts and held that the signs attributed to the mornings, the masculine ones, are those from the ascendant to the middle of the sky and those opposite them, from the setting point to the cardine of the earth; and they held that the two remaining quarters are attributed to the evenings and are feminine, so they made the signs that fall in these two quarters feminine and attributed them to the evenings.

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They have given the signs many epithets, taken from the figures to which they are attributed: they called some of them four-footed, some wild, some governing, domestic, of much sowing, of little sowing and the like. They said that the domestic signs are Aries and Taurus; the wild, Leo and Sagittarius; the speaking signs, which resemble human beings, Gemini, Aquarius, Libra, Virgo and the first half of Sagittarius; the winged, Virgo, Sagittarius, Gemini and Pisces; the *maʿqūma*, Aries, Taurus, Scorpio, Capricorn and Cancer; those of few offspring and little sowing, Aries, Taurus, Libra, Sagittarius, Capricorn and Aquarius; those of many offspring, Cancer, Scorpio and Pisces; the barren, Gemini, Leo and Virgo; those with mutilated limbs, Aries, Taurus, Leo and Scorpio; those of much anger, Aries and Leo; those that signify beauty and comeliness, Gemini, Libra, Virgo, Scorpio, Sagittarius and Pisces; those that gather and fill, Aries and its triplicity; those that give great wealth, Capricorn and its triplicity; those that pour out and empty, Libra and its triplicity; and those that grasp and take, Cancer and its triplicity. The signs of a strong voice are Gemini, Virgo and Libra; the moderate, which have half a voice, Aries, Taurus, Leo and Sagittarius; those of a weak voice, Capricorn and Aquarius; and those with no voice, Cancer, Pisces and Scorpio. Those that signify great care are Leo, Scorpio and Capricorn; those of little care, Virgo and Libra, for their darkness (?); and those of cunning and deceit, Leo, Sagittarius, Capricorn and Pisces. And they said that Aries, Taurus and Gemini, when their lords are found in them, ⟨are⟩ moving; Cancer, Leo and Virgo, when their lords are found [in them], are at rest;

[9] *Tetrabiblos* I.13–17.

[10] Ptolemy explains why only these distances are used in *Tetrabiblos* I.13, al-Battānī in chapter 54 of the *Zīj*. «The two maleficent planets» reads النحسين for the transcription's الجنسين: the houses of Saturn, Capricorn and Aquarius, are opposite those of the moon and the sun, Cancer and Leo, and the houses of Mars, Aries and Scorpio, in quadrature to them (*Tetrabiblos* I.17).

[11] *Tetrabiblos* I.12. Ptolemy declines to list the epithets that al-Battānī gives below.

Libra, Scorpio and Sagittarius likewise, when their lords are found in them, are moving; and Capricorn, Aquarius and Pisces likewise, when their lords are found in them, are at rest.[12]

## ⟨8⟩ On the planets' becoming masculine and feminine relative to the sun, and also relative to the horizon[13]

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They hold that the planets become masculine and feminine according to the configurations they have relative to the sun: when they are eastern, preceding the sun, they are masculine, and when they are western, following it, they are feminine. This also happens relative to the horizon: when they are in the positions from the east to the middle of the sky, or from the west to the cardine of the earth, they are masculine, because these two quarters are masculine; the first of them is called «the swift» and the second «the middle of speed». When they are between the middle of the sky and the west, or between the cardine of the earth and the ascendant, they are feminine, because these two quarters are feminine; the first of them is called «the middle of slowness» (?), and the second, between the ascendant and the cardine of the earth, «the slow and heavy».[14]

## ⟨9⟩ On facing, illumination and the like[15]

It is said that the planets have a proper facing when each of them shares in a configuration with the sun and the moon in this way: that there is between it and the sun or the moon the same distance as between its house and the house of the sun or of the moon, except that it is western of the sun, I mean in the following signs, and eastern of the moon, in the preceding signs. For example, Venus in sextile to one of the luminaries, but in the third sign from the sign of the sun in the order of the signs, or in the eleventh sign from the sign in which the moon is; or Mercury in the second sign from the sign of the sun, or in the twelfth from the sign of the moon. For the other planets reckon their houses in the same way. When the planets are in this condition with respect to the luminaries, they are facing them.[16]

As for illumination, it is said [to be] when the planets are in their own domain and in places like it that agree with them; and when one of them has two or more of its dignities, as when it is in its house, its exaltation, its triplicity and its term and the like, [it is said] to be in the place of its joy and honour, and then its power increases in its actions that resemble the signs that belong to it and are akin to them in nature. It is also said that they rejoice when they are in the dignities of planets of their own domain; this is weaker than the first, because it is below it in power, but they share in it in this way all the same: just as, when they are found in foreign places or in a domain contrary to their own, their proper power weakens, and the contrariety of their mixture to the mixture of the signs in which they are becomes a cause for an action compounded of the two different natures to arise from them.[17]

[12] The meaning of المعقومة, *maʿqūma*, from the root of barrenness, is not clear; the barren signs follow under their usual name, العقيمة.

[13] *Tetrabiblos* I.6.

[14] The words from «they are masculine, because» to «or between the cardine of the earth» are written in the margin. «The swift» reads السريع for الشريع, and «the slow» البطيء for البطن, «the belly».

[15] *Tetrabiblos* I.23, where Robbins translates Ptolemy's terms «proper face» and «chariots».

[16] The manuscript first wrote «or in the twelfth from the sign of the moon, and the other planets», struck it out, and wrote the correct passage in the margin, from «in the order of the signs» to «or in the twelfth».

[17] «Domain», حيّز, *ḥayyiz*; Ptolemy speaks here of the «sect» of the planets.

## ⟨10⟩ On the familiarity of the triplicities with the countries and the planets[18]

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Ptolemy divided the whole inhabited earth that lies to the north of the equator by two lines, which cut it in length and in breadth into quarters, and made each quarter akin to the triplicity, among the triplicities of those four [sets of] signs, to which it is attributed. Ptolemy made the parts of each quarter that lie next to the middle of the whole inhabited earth receive familiarity from the triplicity that governs the angle opposite to that angle, and made Mercury a partner with the governing planets, because it is of a common domain; and made the familiarity of the countries to be with the planets that are the lords of the houses of those triplicities, together with the signs of the triplicities too. Each of these quarters Ptolemy divided into three parts, by which the countries in it are marked off, so that each of these three parts became akin to one sign of the triplicity, together with the planet that is the lord of that sign. Ptolemy explained the natures of the nations that live in each of these particular countries, their characters and their conditions, and treated each of the quarters as a whole according to what belongs to it, the triplicities akin to it and the planets that govern those triplicities, and according to the natures of the signs and of the planets that are their lords, as was needed, and summed it up in what we can dispense with mentioning here; then described the familiarity of each of the countries in each quarter, and of what in it lies next to the middle of the inhabited earth, and classified it separately by the signs and the planets that are the lords of those signs, so that it might be easy to consult, following what had been said before. We have confined ourselves to that and set it out in this way: we began with the countries akin to each sign and its lord, attributed to that third, and followed it with the countries that lie next to the middle of the inhabited world, and we set out in detail what makes each of them known. Then we recorded the familiarity of the cities with the signs and the planets, according to what is said to have been rising at the time of their building, or the positions of the sun and the moon, or the middle of the sky of the kings who lived at the time when they were built.[19]

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So of the countries and districts, those akin to Aries are the lands of Britain, Bastarnia (?), Germania and Galatia, the land of the Jews and Deep Syria; by sharing with Aries these countries belong to Mars. Those akin to Taurus are the lands of Parthia, Media, Persia, the Cyclades, the coasts of Asia Minor and Cyprus; by their familiarity with Taurus they belong to Venus. Those akin to Gemini are the lands of Jurjān, Ṭabaristān, Armenia, al-Qayrawān, Marmarica, which is below the land of Egypt, and «فورينا»; by their familiarity with Gemini they belong to Mercury. Those akin to Cancer are Numidia, Carthage (?), Ifrīqiya, «سوسا» and Phrygia; by their familiarity with Cancer they belong to the moon. Those akin to Leo are Italy, «وعاليلوابولا», Sicily, Phoenicia (?) and Chaldaea (?); by their familiarity with Leo they belong to the sun. Those akin to Virgo are Babylon, the Jazīra, the land of Āthūr, «الأمن», Achaia (?) and Crete; and they are akin to Mercury too. Those akin to Libra are «ينظر بالي», Casperia (?), Serica, Thebais (?), the Oasis (?) and Troglodytica (?); and they are akin to Venus too, with Libra. Those akin to Scorpio are Metagonitis (?), Mauritania (?), Gaetulia (?), Syria, Commagene (?), «فافا حرفا» and the Ḥijāz; and they are akin to Mars too. Those akin to Sagittarius are Tyrrhenia (?), Celtica (?), Spain (?) and also the cultivated land of the Arabs; their familiarity is with Jupiter. Those akin to Capricorn are the land of India, Ariana, «ادنا», China, Gedrosia (?), Thrace (?), Macedonia and Illyria (?); and they are akin to Saturn. Those akin to Aquarius are Sarmatica (?), «طلساي», Sogdiana (?), «ارانيانا» and the Middle «سوبا», meaning the middle land of the Abyssinians; and they are akin to Saturn too. Those akin to Pisces are Phazania (?), «فعلاعرسا», Nasamonitis (?), Garamantica (?), Lydia (?), Cilicia (?) and Pamphylia (?); and they are akin to Jupiter, with Pisces. All of these are «ما يميّزه» and ninety regions, of those in the four [quarters] mentioned and of those next to the middle of

[18] *Tetrabiblos* II.3. In the lists of this chapter, names that the Arabic forms allow to be identified are given in English, and «(?)» marks a doubtful identification; names too corrupt to identify are given in Arabic letters as the transcription has them.

[19] «Inhabited» reads المعمورة for المحمودة, «praised», and «the cities» المدن for الدرج, «the degrees».

the inhabited world.[20]

One must add to this as well that each of the fixed stars is akin to the countries to which the parts of those signs are akin, and they all act on them: these are the parts that the stars face and that correspond to them on one of the circles that pass through the poles of those signs; this is evident from the latitudes of the stars when their passage is above the zenith. This is the account of what we have described concerning the cities.

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The cities of the Ḥijāz: Scorpio. Mecca: Libra and Saturn. The City of the Messenger, on whom be peace: Scorpio and Venus. Al-Rabadha: Taurus and the moon. Al-Ṯaʿlabiyya (?): Sagittarius and Mercury. The cities of Yemen: Sagittarius. Ṣanʿāʾ: Gemini and Mercury. Al-Janad (?): Pisces and Mars. ʿUmān: Taurus and Venus.[21]

The cities of Iraq: Leo. Basra: Pisces and Mercury. Kufa: Virgo and Jupiter. Baghdad: Taurus and Mercury. Wāsiṭ: Pisces and Jupiter. Daskarat al-Malik: Leo and Jupiter. Al-Madāʾin: Leo and Jupiter. Al-Sūs: the Head and the moon.

The cities of Māh: Taurus. Ḥulwān: Taurus and the moon. Al-Dīnawar: the Head and the sun. Nahāwand: Scorpio and Mars. Hamadhān: Taurus and Venus. Al-Rayy (?): Taurus and Saturn. Nahāwand: the Head and the moon. Al-Daylam: Scorpio and the moon. Ṭabaristān: Capricorn and the moon. Sāriya of Ṭabaristān: Aries and Venus. The city of Jurjān: the Head and Jupiter. Qūmis: Capricorn and the moon. Iṣfahān: the Head and the sun. Nīsābūr: Aries and Jupiter. Nasā: the Head and the sun. Abīward: Libra and Mercury. Ṭūs: Aries and Mercury. Sarakhs: Virgo and the moon. Marw al-Rūdh: Gemini and Saturn. Al-Ṭālaqān: Libra and Mars. Marw: Pisces and Saturn. Balkh: Libra and Mars. «أحل» of Khurāsān: Taurus and the moon. Khwārazm: Gemini and Saturn. Samarqand: Aries and Mercury. «الدسقوته»: Libra and Mars. Al-Shāsh: Virgo and the sun. Farghāna: Taurus and Jupiter. Qirmīsīn (?): Gemini and Venus. «يهله»: Cancer and Saturn. Bādghīs (?): Gemini and Mercury. Sijistān: Cancer and Jupiter. Kābul: Aries and Jupiter. Al-Dāwar (?): Gemini and Scorpio. Zābulistān: Aries and the moon. The city of Tibet: Capricorn and the sun. Makrān: Gemini and Saturn.[22]

The cities of India: Capricorn. «شامل»: Libra and Capricorn. «قودين»: Sagittarius and the moon. «مالود»: Capricorn and Mars. «الهند مار»: Gemini and the moon. «قارورة»: Pisces and the sun. «طبرية»: Aries and Mars. «طعمة»: Capricorn and Jupiter. The cities of China: «اسعير»: Leo and the moon. «كسعير»: Taurus and the moon. The city of China: Leo and Venus.

The cities of Āthūr, which is the country of Mosul: Virgo. Nīnawā: Capricorn and Jupiter. Irbil: Scorpio and Jupiter. «الحومة»: Scorpio and the moon. «برسيار»: Pisces and Venus. «أليس»: the Head and Saturn. Takrīt: Virgo and Saturn. Al-Karkh: Libra and Mars.

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The cities of Ādharbayjān: Ardabīl: Taurus and Venus. «بهدى سابور»: Gemini and Mercury. Fasā: Libra and Venus. The city of Kirmān: Libra and Mars. The Kirmān that is «سرّ» Kirmān: Aquarius

[20] Ptolemy's list (*Tetrabiblos* II.3, at the end): Aries, Britain, Gaul, Germania, Bastarnia, and in the centre Coele Syria, Palestine, Idumaea, Judaea; Taurus, Parthia, Media, Persia, and the Cyclades, Cyprus, the coast of Asia Minor; Gemini, Hyrcania, Armenia, Matiana, and Cyrenaica, Marmarica, Lower Egypt; Cancer, Numidia, Carthage, Africa, and Bithynia, Phrygia, Colchica; Leo, Italy, Cisalpine Gaul, Sicily, Apulia, and Phoenicia, Chaldaea, Orchenia; Virgo, Mesopotamia, Babylonia, Assyria, and Hellas, Achaia, Crete; Libra, Bactriana, Casperia, Serica, and Thebais, Oasis, Troglodytica; Scorpio, Metagonitis, Mauritania, Gaetulia, and Syria, Commagene, Cappadocia; Sagittarius, Tyrrhenia, Celtica, Spain, and Arabia Felix; Capricorn, India, Ariana, Gedrosia, and Thrace, Macedonia, Illyria; Aquarius, Sauromatica, Oxiana, Sogdiana, and Arabia, Azania, Middle Ethiopia; Pisces, Phazania, Nasamonitis, Garamantica, and Lydia, Cilicia, Pamphylia. The Arabic gives Hyrcania as Jurjān, Cyrenaica as al-Qayrawān, Mesopotamia as the Jazīra and Coele Syria as «Deep Syria». The number at the end is corrupt; the Berlin manuscript has «two hundred».

[21] Ptolemy ends *Tetrabiblos* II.3 by relating each city to the signs that the sun, the moon and the ascendant occupied at its foundation, or to the middle of the sky of its rulers; al-Battānī's list of cities is not in Ptolemy. In it «the Head» (الرأس) stands several times in the place of a sign; whether the ascending node or a sign written in a corrupt form is meant is not clear.

[22] «Al-Rayy» for الرئيسي, «Nasā» for فسا, «Abīward» for أمّا ورد, «Sarakhs» for مرخس and «Marw» for مرد و.

and Mars.

The cities of the Jazīra: Virgo; [then] Scorpio and Mercury. Naṣībīn: Leo and Jupiter. Sinjār: the Head and Jupiter. «قزدين»: Cancer and Jupiter. «كاري»: Libra and Jupiter. Raʾs al-ʿAyn: Aquarius [and] Pisces. Tall Mawzan: Virgo and Saturn. Āmid: Aquarius and the moon. Mayyāfāriqīn: Cancer and Venus. Arzan: Gemini and Mercury. Al-Raqqa: Sagittarius and the moon. Qarqīsiyā: Scorpio and Venus. ʿĀna: Libra and Jupiter. «جيب»: Capricorn and Jupiter. Ḥarrān: Gemini and Mercury. Al-Ruhā: Leo and the sun.[23]

The cities of al-Bāb: the city of Bāb al-Khazar: the Head and Saturn. The city of Ṣūl: the Head and Saturn. Bardhaʿa (?): Aquarius and Saturn. Āmid: Aquarius and the moon. Qabala: Taurus and Mars. «كسل»: Scorpio and the sun. «حزران»: Leo and the sun. Bāb al-Lān: Libra. «راسان»: Aquarius and Saturn. «بارقة»: Libra. «اسيل»: Virgo and Venus. Al-Lān: Pisces and Jupiter. «هيل العين»: Pisces and Jupiter. «رسل»: Virgo and Saturn. «أسري»: Cancer and Mars. «ناليغلا»: Capricorn and Mercury. «ميلان»: Gemini and Mars.

Malaṭya (?): Aquarius and the moon. Qaysāriyya: Pisces and Saturn. Sumaysāṭ (?): Pisces and Mercury. «طراونه» and «بلوانها»: Pisces and Saturn. «قدروفسه»: Capricorn and the moon. Satala (?): Pisces and Saturn. Ṭarsūs: Libra and Mars. Al-Maṣṣīṣa: Leo and the moon. Adhana (?): Sagittarius and Jupiter. «جرجرة»: Taurus. Anqira: the Head and the moon. Salūqiya: Cancer and the sun. «حرىى»: Libra and Mercury. «مويّا»: Aquarius and the moon. Ephesus (?): Scorpio and the moon. Samos (?): Aquarius and Saturn. The island «كاوس»: Cancer and the sun. The island of Rhodes (?): Pisces and Mars. Chalcedon (?): Aquarius and Mercury. Nicomedia (?): Taurus and Saturn. «أو قمر»: Libra and the moon. «سعبه»: Pisces and Venus. ʿAmmūriya (?): Pisces and Venus. Sumaysāṭ: Libra and Mars. The cities of «صوس»: Taurus. «ماسوس»: Virgo and Mars. «امنوس»: Aries and the moon. Constantinople. Cyprus: Taurus and Mercury.

f. 25v

The cities of Syria: Scorpio. Al-Lādhiqiyya: Virgo and the sun. Ṣūr: Gemini and Jupiter. Aṭrābulus: Scorpio and Mercury. Anṭākiya: Aries and Jupiter. «حرقة»: Pisces and Mercury. Manbij: Taurus and the sun. Ḥalab: Scorpio and Mercury. Damascus: Scorpio and the sun. «بيت جدي»: Cancer and the moon. ʿAsqalān: Leo and the sun. Ṣaydā: the Head and the sun. Maʿarrat Maṣrīn: Virgo and Mercury. Tadmur: Scorpio and Mars. Ṭabariyya: Pisces and Venus. Nābulus: Taurus and Venus. Bayt al-Maqdis: Cancer and Mars. «سحره»: Taurus and Jupiter.

The division of «لوبيرومي»: the land of Lower Egypt. «فراس»: Cancer and Mars. Alexandria: Leo and Mars. Memphis (?): Gemini and the sun. ʿAyn Shams: Gemini and Mercury. Aswān. «هذه أدلة»: Libra and Saturn. «ري»: Taurus and the moon. «مرسله»: Scorpio and Mercury. Armant (?): Gemini and Jupiter.

The cities of «البريدة»: Cancer. «قرطبة»: Taurus and Jupiter. «سنما»: Gemini and the moon. The cities of Pentapolis: Gemini. Ifrīqiya: Aries and the moon. Arsinoe (?): Aquarius and Saturn. Ptolemais (?): Taurus and Jupiter. Apollonia: Pisces and the sun. «فروس»: Pisces and the moon. Marāqiya: Cancer and the sun. The cities of Ifrīqiya: Cancer. «سرّ»: Taurus and Mercury. «سرطوبة»: Cancer and the sun. Carthage (?): the Head and Mars. Great Leptis (?): Aries and Jupiter. «سفر»: Leo and the moon. Al-Qayrawān: Scorpio and the sun. The cities of Ṭanja: Scorpio. The city of Ṭanja: Aquarius and Saturn. Caesarea of Ṭanja: Pisces and Venus. Walīla (?): Aries and the moon. «لو يحين»: Cancer and Jupiter. Al-Sūs of Ṭanja: Virgo and the moon. «روحا»: Cancer and Jupiter.

The division of the regions of Constantinople: the City of the King: Cancer and Mars. Great Rome: Leo and the sun. The island of «يول»: Aries and Mercury. The island of «ابيرسا»: Scorpio and Saturn. «باي»: the Head and Mars.

[23] «Jupiter» reads المشتري for المشاري, and «ʿĀna» عانة for غانة. After «Virgo» the name of a city seems to be lost.

The cities of al-Andalus: Sagittarius. «باريا قيساريا»: Virgo and Mars. «إيحاء وسطًا»: Pisces and the sun. Hispalis (?): Gemini and Mercury. Qurṭuba: Cancer and Jupiter. Asturia (?) [and] «ارغسطا»: Leo and Jupiter. Ṭarrakūna: Scorpio and the moon. «قلوبيّا»: Leo and the moon. «ما رينيون»: Virgo and Saturn. The island «كريبا»: Cancer and Venus. Massilia (?): Libra and Mercury. «بابوسا»: Libra and the moon.[24]

f. 26r

The cities of Germania: Aries. «لوقة»: Gemini and Mars. «دباورا»: Virgo and Mars. «سهند»: Capricorn and Jupiter. «بعطليون»: Scorpio and Saturn. «إزالاطى»: Aries and the sun. «اابا»: Pisces and Mars. «سوبر طيون»: Aries and Mars. «لابارزطا»: Aries and Saturn. Sirmium (?): Capricorn and Mars. «سندون»: Virgo and Mars.

The cities of Italy, the first of which is Rome, which we have mentioned: Scorpio and Saturn. «طراقه»: Aries and Mars. «ماقلس»: Taurus and Venus. «رأسا»: the Head and the sun. «بليدس»: the Head and Saturn.[25]

The cities of the island «ورأس الأدنى»: Gemini and the moon. «هاريني»: Virgo and the moon. The cities of the island of Sardinia: «سالقو»: Cancer and Saturn. «واداس»: Taurus and Venus. «طورواس»: Pisces and Mars. The cities of the island of Sicily: «لسسدي»: Leo and the moon. «اعا»: Cancer and Jupiter. «سوس»: Aquarius and Jupiter. «طرسقي»: Aquarius and Mercury. Syracuse (?): Pisces and Mercury. Catania (?): Scorpio and Jupiter. «باجور»: Capricorn and Saturn. «أسوس»: Taurus and Saturn. «لاوس»: Leo and Mars. «ليوفيتا»: Capricorn and Mercury. «واربيوس»: Scorpio and Mercury. «يرسما ينسون»: Capricorn and Mercury. «ديراقولسوس»: Cancer and the moon. «رينون»: Leo and Venus. Thessalonica (?): Leo and the sun. «بالاهي», the city of Alexandria: Taurus and Mercury. «ولارينا»: Virgo and Mercury. «قساير»: Scorpio and the moon. Thebes (?): Taurus and Mercury. «فارقور»: Virgo, Mercury and Jupiter. «نفافالاما»: Pisces and Saturn.

f. 26v

In this copy of the cities we found the familiarity of the districts with the signs in the form that I shall set down. You see that this copy is one whose interpretation was taken care of, and that it seems to differ from Ptolemy in some districts and to lean in part to the opinion of the Egyptians in the division of the countries. Of the districts, those akin to Aries are Fārs, Ādharbayjān, Britain, «طعنه», Germania, Palestine and part of al-Balqāʾ; and they are akin, with Aries, to Mars. Those akin to Taurus are all the land of Māh, Iṣfahān, the small islands of the Sea of the Romans, Cyprus and «أثار»; they are akin, with Taurus, to Venus. Akin to Gemini are «حلان», al-Daylam, Jurjān, Ṭabaristān, Greater Armenia, «مرم», Pentapolis, Marāqiya and Egypt; and with Gemini, Mercury. Akin to Cancer are the land of the Berbers, Ifrīqiya, «وينبه إلى» the land of the Romans, Phrygia (?), «لاوقة» and «لاودية»; akin with Cancer is the moon. Akin to Leo are the land of the Turks, «سمر», Italy (?), Cilicia (?), «انقبيه», the land of Ḥimṣ, Damascus and the Sawād of Kufa; and with Leo, the sun. Akin to Virgo are the land of Babylon, «مواقيّة», the land of Mosul, the Jazīra, the land of «الرياس» and Crete; and with Virgo, Mercury. Akin to Libra are the land of Bukhārā⟨n⟩, «حارستان», Ḥimyar, Tibet, the island of Sardinia (?), a part of the land of the Abyssinians that is called «طراموية», Ṣanʿāʾ, the land of Najrān and the land of Kirmān; and with Libra, Venus. Those akin to Scorpio are the country of the Ḥijāz, «امد», Ṭanja «بأطالية», «ارزايّا», al-Samāwa (?), «سريرا», Cappadocia (?) and «فلسعه»; and with Scorpio, Mars. Akin to Sagittarius are «ملطيقي», al-Andalus, Yemen and the island of Sicily, which is in the Sea of Syria; and with Sagittarius, Jupiter. Akin to Capricorn are India, al-Sūs, «يكران», Harāt, «براقيّة», «سقالية» and «أنيلوس»; and with Capricorn, Saturn. Akin to Aquarius are «سردا ونطيقي», which is Balkh, «الصغير», Farghāna, «انيما» of «البلقى», and the middle of «الجمش»; and with Aquarius, Saturn. Those akin to Pisces are

[24] «Gemini» reads للتوأمين for لليومان, and «the moon» القمر for المقر.

[25] «Italy» reads إيطالية for أنطالية.

«قاتومه», «بطلموسه», «بسهوطس», «حرفه» and «نبد لوقية»; and with Pisces, Jupiter.[26]

## ⟨11⟩ On knowing the events that occur in the countries from eclipses[27]

f. 27r

We begin by examining the place of the eclipse in the sphere of the signs: to which of the triplicities it belongs, and to which of the regions that triplicity is attributed. Then we turn to the countries and cities that are akin to [the sign of] the eclipse in particular, in that triplicity, as we have set out, and see whether the eclipse is seen in all those places or in some of them; for in those of them where the eclipse [does not] appear nothing happens. We set up the ascendant of the middle of the eclipse and its cardines for the places among those countries where we know that it is seen, reckoning the hours and the altitude of the pole there. After that we infer the time of the event and the length of its duration by seeing how many equal hours there are from the beginning of the darkening of the eclipse to the end of the clearing in each of those places. If the eclipse is solar, the event lasts as many years as there are hours; if it is lunar, as many months. If the place of the eclipse is on the eastern side of the horizon, we know that the event begins in the first third of that period and that its greatest intensity falls in that first third; if it is on the side of the middle of the sky, we infer that it is in the second third of the period; and if on the western side of the horizon, in its last third. We determine the place of the eclipse relative to the horizon as I shall describe to you. If the place of the eclipse is less than 4 seasonal hours from the ascendant towards the middle of the sky, it is on the eastern side; if it is past the 4 hours, up to the cardine of the sky and 2 seasonal hours beyond it, it is on the side of the middle of the sky; and if it is past that, it is on the western side. In this you proceed on the basis that if it is in the very degree of the ascendant, [the event] is at the beginning of the period; if it is in the degree of the middle of the sky, in the middle of the period; and if it is in the very degree of the setting point, the event is at the end of the period.[28]

The particular increases and decreases in the events are known from the conjunctions and oppositions that occur in the meantime, if they happen in the places that are the cause of the events or in places that share in a configuration with them, I mean the places of the eclipse and of the cardines and the places that share in a configuration with them; and from the positions of the very wandering stars that are in charge of the governing, when they appear eastern or western, or stationary, or rising at the beginning of the night, and share in a configuration with the signs of the eclipse or the signs of the cardines, which are the cause of the events. For if they are eastern or stationary, an increase of the events comes from them; and if they are western, or under the rays, and rising at the beginning of the night [and] retrograde, a decrease of the events comes from them. Look also at the decrease of the upper planets and their staying in each of the times, I mean the stations of Saturn, Jupiter and Mars, when they are stationary and are the lords of the indication through their dominion over the places of the eclipses and of the cardines, in the way that we shall mention.

f. 27v

The class in which the event occurs is known from the conditions and the forms proper to the signs in which the eclipses occur, and in which are the wandering and the non-wandering stars that govern the sign of the eclipse; and from the governing star, and the sign of the cardine that precedes the eclipse. This cardine, which the eclipsed luminary passes before the time of the eclipse, is either the ascending cardine or the cardine of the middle of the sky, whichever it happens to pass before the eclipse. The governor of these two places among the wandering stars is known as I shall describe to you. Look for the planet that has many shares in these two signs that we have mentioned: either it is conjoined with the degree of the eclipse or the degree of the cardine, or is coming to it, or has just left it (?); or it shares in a configuration with it by

[26] The copyist repeated «and with Sagittarius, Jupiter» and struck the repetition out.

[27] *Tetrabiblos* II.4–8.

[28] «[Does not]» supplies a negation that the sense requires: Ptolemy considers the countries where the eclipse is visible (*Tetrabiblos* II.5–6). The division of the sky by seasonal hours is al-Battānī's addition.

one of the connections, I mean the trine, the sextile, the quadrature, the opposition and the like; [and look] at the dignities it has in the signs, by being the lord of the house, of the exaltation, of the triplicity and of the terms. If all this comes together in one planet, it alone is the lord of the governing. If no single planet is found that is the lord both of the eclipse and of the cardine, one should take together the two planets that have the most familiarities with each of the two places, as said before, choosing first the sign of the eclipse and putting it first. If the planets that have shares in these two places are more than two, and what each of them has in the two places is equal to what the other has, we choose for the governing, and put first, the one of them that has the better claim to be said to be in the cardine, or is stronger in action, or is stronger in that domain.

f. 28r

We know the governors among the fixed stars as well. We begin with the star that was with the place of the eclipse in the cardine that came before the eclipse: we mean, among the bright stars, the star that rose with the degree of the eclipse or culminated with the degree of the eclipse, in whichever of these two cardines one of the luminaries passed before its eclipse — we mean the great stars — [...] on people and on kings. Likewise, if at the time of the eclipse they are in the places that Ptolemy described, in the east in particular, they signify that the event will be in the fruits, in the age of youth, in foundations and in whoever is young in age; if they are in the middle of the sky above the earth, they signify that the event will be in the temples, in the kings and in those of middle age; and if they are in the west, they signify that the event will be in the matter of the customs and their changes, and in those of old age, and the dead (?).[29]

The size of the part of the class concerned in which the event occurs is known from the amount of the darkening of the eclipse and from the relation of the planets in charge of the governing to the place of the eclipses. When they are in configurations attributed to the evenings at eclipses of the sun, or in configurations attributed to the mornings at eclipses of the moon, the events are in the smaller part of that class; if they are in opposition, on the diameter, the event concerns half of that class; and if they are in configurations attributed to the mornings at eclipses of the sun, or attributed to the evenings at eclipses of the moon, it is in the greater part of that class. Whether what happens is good or the opposite is known from the nature of the governing, chief planets that bring about the event, from their mixture with one another and from the places in which they are. For the sun and the moon are two princes and chiefs of the other stars, in that they are the cause of all the action, the cause of the governing of the stars, and the cause of the strength and weakness of the governing stars.

f. 28v

The consideration of the mixture of the planets in charge of the governing with one another is what shows the condition of the events that occur. Saturn, when it alone is the lord of the governing, is in general the cause of corruption that comes from cold. As for its particular action proper to it: if the event is in people, it produces long illnesses, consumption, wasting, harm from humours and the flowing of superfluities, quartan fevers, flight, perplexity, imprisonment, grief, fear and death, especially for those advanced in age. If the event is in the beasts that people use much, the harm to them is their scarcity: corruption of their bodies and diseases befall those of them that exist, so that, because of that, something like these illnesses befalls the people who eat these beasts. If the event is in the air, there comes severe cold with frost and fog, heat (?) that brings pestilence, badness of the air, thickness of the clouds, gloom and destructive snowstorms, and from this many vermin are generated that are harmful to the nature of people. If the indication is in the rivers and the seas, there comes, in what they have in common, wintry weather, the sinking of ships, difficulty in travelling on the water, and scarcity and corruption of fish; in the seas in particular ebb and flood, and in the rivers in particular excessive rising and badness of the waters. If the events are in the fruits that grow from the earth, they suffer want,

[29] Something seems to be lost before «on people and on kings»: Ptolemy's account of the classes signified by the forms of the signs ends with the bicorporeal signs, which concern people and kings (*Tetrabiblos* II.7). «Customs» reads السنن for السنين, «the years».

scarcity and corruption, especially those that people need of necessity, and the corruption that befalls them comes from caterpillars or from locusts, or from floods, or from heavy rains, or from hail and the like, so that at times drought comes of it, and people perish because of it.

Jupiter, when it alone is the lord of the governing, in general makes things grow. Its particular actions in what happens are, in people, elevation of rank, good condition, plenty, a state of safety and security, the growth of necessary things, the good condition of bodies and souls, rewards and gifts from kings, the increase of the affairs of kings, their greatness and their magnanimity; in general it is the cause of good fortunes. In beasts, if the event is in them, those that people use much multiply and increase greatly, and those of the opposite kind decay and perish. As for its indication in the air, [the air] is temperate, healthful, windy and moist, and it makes the fruits that grow from the earth increase; the passage of ships at sea is safe, the risings of rivers are moderate, and all things like these. Whenever its indication of the event is in one of the kinds named, it sets it right, and what comes of it is good, moderate, agreeable, praiseworthy and conducive to well-being.

f. 29r

Mars, when it alone is the lord of the governing, is in general the cause of the corruption that comes from dryness. Its particular actions: if the event is in people, wars arise among them, and seditions between a people and those of their own kind, captivity and slavery, the uprising of the mass of the people and the wrath of the leaders; from these things sudden death befalls some people, and illnesses with tertian fevers and the spitting of blood, and swift and violent death, especially for those in the age of the prime. Likewise some suffer robbery, contempt, aggression, lawlessness, arson, the killing of men, plunder and highway robbery. If the event is in the air, there come severe heat, hot, pestilential and harmful winds, the falling of thunderbolts and of the thunderbolt that is called *prēstēr*, and scarcity of rain. If it is at sea, there comes the sudden sinking of ships through changeable and turbulent winds, through thunderbolts and the like, thunder, burning and what resembles them. In the rivers there come a decrease of the water, the drying up of springs and the corruption of drinking water. In the animals and plants that are fit for the use of people there comes scarcity, and the corruption of the fruits, which is produced by burning «احرًّا ناها», or by the pestilence of locusts, or by the scattering that the winds bring upon them, or by their burning in the stores in which they are kept.[30]

Venus in general produces something like what Jupiter produces, and in what comes from it there is, besides, a stirring of matters of intercourse, of amusement and of the affairs of women. Its particular action, in people, is elevation of rank, joy, good condition, the soundness of marriage, many children, the success of all matters of union, the increase of possessions, clean and pious conduct, goodness of conduct in matters of religion, health of the bodies, connection with leaders, and marriage alliances with governors. In the condition of the air temperate, moist and very nourishing winds come, and the air is good and clear; the fertilizing waters and the rains it makes abundant; ships come through safely at sea; and from it come successes in trade, profits, and the rising and fullness of the rivers. The animals that people use, and the fruits of the earth in particular, increase, flourish and are of use; and whenever the event falls in any kind of these kinds, its condition is praiseworthy and useful to people.

f. 29v

Mercury, when it alone is the lord of the governing: in general, when it mixes with any of the other planets, it is akin to their natures (?). Its proper action is to set things in motion, more than the other planets. Its action in the events concerning people is sharp, strong and subtle in contriving the thing it aims at; it is a maker of highway robbery, thefts, brigandage and raids, and it also causes difficulty in sea voyages when it shares in a configuration with the maleficent planets. It is the cause of dry illnesses, of fevers that come every day, of coughing, spitting and consumption; and [the cause] of the matters of divination, of the worship of God, the blessed and exalted, of what is levied on the revenues of the temples, and of the change of

[30] *Prēstēr* renders فريطس, the Greek πρηστήρ, a fiery whirlwind; Robbins translates «hurricanes».

laws and customs at some times, in proportion to its mixing with each of the planets. Since its mixture is dry and quick in motion, because of its nearness to the sun and the speed of its return in its revolutions, it produces in the air in particular turbulent, swift and easily changing winds, thunder, the thunderbolt that is called *prēstēr*, the sinking of the ground, earthquakes and lightning; and sometimes, because of these things I have mentioned, corruption comes to the animals and plants that people use much. At the times of its setting it causes a decrease in the waters and the rivers, and at the times of its rising it increases them.

f. 30r

Each of these planets, when it is in the natural condition proper to it, produces things such as these. When they are mixed with one another, according to their sharings in the configurations, the diversity of the signs, their appearances relative to the sun, and their separations and meetings, then, from the mixing that there is in their actions too, their proper action becomes of different kinds accordingly, and mixed from the natures that have shared in its nature. Since the description of the proper action that comes from each kind of mixture is without end and impossible, and, in general, the mention of all that, in whatever configurations it happens to be, may be understood in many ways, we must leave this kind to the one who considers it, so that they may examine the particulars in it. One must look at the familiarity of the countries and cities with the planets from which the events that befall them come.

For if those planets are beneficial and akin to the places in which the events occur, and the maleficent planets contrary to them do not overpower them, their action proper to their natures is more beneficial; just as, if they are not akin to those countries and the maleficent planets overpower them, their benefit is less. If their mixture is maleficent and harmful and they are the agents of the event, then if they are akin to the countries in which the events occur and the beneficent planets overpower them, their harm is slight; and if they are not akin to the countries and the beneficent planets do not overpower them, the effect of their corrupting mixture on them is greater.

Ptolemy's words «the planets overpower them» admit many meanings and senses. One of them: a planet is in conjunction with a planet; then the one of the two with the greater latitude in the north overpowers the other; if both are in the south, the one with the smaller latitude overpowers; and if one is in the north and the other in the south, the northern overpowers the one in the south. Another: a planet is akin to a planet by the configuration of quadrature; then the one in the preceding signs overpowers, because it is in the tenth sign from the sign in which the other is, which is the middle of the sky of that sign. Another: the overpowering is in the way that we mentioned at the beginning of the book, so that one of them is in the high signs and the other in the low signs, and their distances from the two equinoctial points are one and the same. Another: one of them is in a cardine and the other declining from the cardine, or one of them is above the earth and the other below the earth. In general, when the beneficent planets look at the maleficent, from whatever configuration it may be, they diminish their evil; and when the maleficent look at the beneficent, they diminish their benefit, especially if it is from quadrature and opposition. What Ptolemy mostly means by overpowering is that a planet follows a planet and rises to its place: that is, when it rises after it, so that the first declines and the other remains in its place and overpowers it.[31]

f. 30v

For the most part the general accidents befall people when [those places] are the necessary places of their nativities, I mean the places in which the luminaries are, or the places of the cardines, in the nativities; for these are the places of the things that are the causes of the general accidents, whose causes are the eclipses occurring in those places or in the places opposite them. It is most severe and hardest to guard against when the degrees of the place of the eclipses occurring in those places, or in the places opposite them, are the degrees of the places of the luminaries in

[31] The high and the low signs at equal distances from the equinoctial points are the commanding and the obeying signs of chapter 6 (*Tetrabiblos* I.14).

the nativity, and next the degrees opposite them. The most severe of them is an eclipse of the sun, because it can only occur at the revolution of the year, while the moon may be eclipsed in its own place in the root [nativity] at the time of the revolution.

### ⟨12⟩ On the colours of eclipses and the stars with tails[32]

One should also consider the colours that appear at the times of eclipses, I mean the colours of the luminaries themselves and the colours of the things around them, when such arise around them, like those called rods, staffs, haloes and the like. If they are black, dense and inclining to green, they signify the things that we have said are of the nature of Saturn; if white, the things of the nature of Jupiter; if inclining to red, the things of the nature of Mars; if tawny and golden, the things of the nature of Venus; and if of varied colours, the things of the nature of Mercury. If the colour is over the whole body of the luminaries, or in all the places around them, the event will be in most parts of the countries; if it is in one part of the luminaries or of what is around them, whatever part it may be, the event befalls only the region that faces the part in which the colour is.

f. 31r

You should also look at the stars with tails that appear at the times of eclipses or at other times, and at the other general phenomena that arise, like those called staffs, trumpets, jars and the like; and take their natures, in what they produce, to be the same as what we said the nature of Mars and the nature of Mercury produce: wars, burning, severe heat, earthquakes and the accidents that come from them. The places in which the events occur are inferred from their positions in the sphere of the signs, or from the places that the direction of the tail of the stars faces. From the form of the phenomenon that appears one infers the kind of event that occurs; the class in which the event occurs, from the positions in the sphere of the signs in which that thing appears; the duration of the events, from the time for which the phenomenon that appears lasts; and the time of the beginning of the event, from its relation to the sun. For for the most part, when they appear while attributed to the mornings, the event from them is swift, and when they appear while attributed to the evenings, the event from them is slow.[33]

### ⟨13⟩ On knowing the conditions of the year[34]

In determining the year we should use the entry of the sun into the beginnings of the solstitial and equinoctial signs, I mean the points of the two equinoxes and the two solstices, which are the beginnings of the seasons of the year; and examine the conjunctions and oppositions of the sun and the moon that occur near these beginnings, before these times, and especially those of them at which an eclipse occurs. So from the beginning [taken] at the sun's entry into Aries it may be known what the spring will be like; from the beginning found at its entry into Libra, what the autumn will be like; from the beginning found at its entry into Cancer, what the summer will be like; and from its entry into Capricorn, what the winter will be like; since these general times and their conditions come only from the sun.[35]

f. 31v

In this we should use the properties of the signs, and infer from them the winds that blow from the quarters and their natures in general. The changes towards increase and decrease at some times are also inferred, in general, from the conjunctions and oppositions that occur after the sun's entry into the beginnings of these signs, that is, the conjunctions and oppositions that follow these points, through the wandering stars' sharing in configurations with them. The particular

[32] *Tetrabiblos* II.9.

[33] «Like those called» follows the Berlin manuscript and the Arabic *Tetrabiblos* (مثل); for the jars (T: الخوابئ) the transcription has والحرّاي. The words from «to the mornings» to «while attributed» are written in the margin.

[34] *Tetrabiblos* II.10–12.

[35] «Examine» follows the Berlin manuscript (تتفقّد); the transcription has تتعقد.

matters that occur in one of the quarters [of the year] are inferred from the conjunctions that occur in each of the signs and from the positions of the planets; and this may be called the examination of the months. In this subject, besides what we have said, one needs to know the particular natural properties of each of the signs as they bear on the conditions of the year. The mixtures of the wandering stars, and of the fixed stars whose mixtures are like those of the wandering ones, and their familiarity with the air and the winds, we have mentioned before, and likewise the familiarity of all the signs with the winds; it remains only to report their particular natures and their familiarity with the seasons, I mean the familiarity of each sign.

f. 32r

Aries is on the circle of the equator, and the nature of the sign of Aries as a whole, because of the equinox that is in it, generates thunder and lightning; its northern part is hot and corrupting, and its southern part generates frost and is cold. Taurus is nearer to the summer circle; the sign of Taurus as a whole signifies both mixtures, inclining to heat; its northern part is of a temperate mixture, and its southern part generates disturbance and earthquakes. Gemini is nearer to the summer circle; its nature as a whole produces a temperate mixture; its northern part generates winds and produces earthquakes, and its southern part is fiery and burning. Cancer is on the summer circle; as a whole it produces fair weather and is hot, and its northern and southern parts are fiery and burning. Leo is nearer to the summer circle; as a whole it is burning, with hot, stagnant air and pestilence; its northern part is stirring and fiery — that is, it stirs up hot winds — and its southern part moistens. Virgo is nearer to the circle of the equinox; as a whole it moistens and produces thunder; its northern part produces winds, and its southern part is of a temperate mixture. Libra is on the circle of the equator; as a whole it is changeable and turning; its northern part generates sand (?), and its southern part moistens and produces pestilence. Scorpio is nearer to the winter circle; its nature as a whole produces thunder and is fiery; its northern part is fiery and burning, and its southern part moistens. Sagittarius is nearer to the winter circle; as a whole it produces winds; its northern part produces winds, and its southern part is very moist and changeable. Capricorn is on the winter circle; as a whole it is very moist, and its northern and southern parts are moist and very corrupting. Aquarius is nearer to the winter circle; as a whole it is cold and watery; its northern part is burning, and its southern part produces burning. Pisces is nearer to the circle of the equator; as a whole it is cold and generates winds; its northern part generates winds, and its southern part is watery.[36]

f. 32v

The nature of the parts of the signs is known from the mixture of the fixed stars in each sign, among the stars in the figures of the sphere of the signs and those outside them, according to their sizes in magnitude and the likeness of their natures to the natures of the wandering stars. Among the nebulous stars in the sphere of the signs, those like the Pleiades, whose mixture is first of the moon and second of Mars, produce earthquakes, winds and fog; the nebulous one in the figure of Cancer, which is called the Manger, whose mixture is first that of Mars and second the moon, produces hot, stagnant air, earthquakes and gloom; the nebulous one that follows the Shawla of the Scorpion, whose mixture is that of the sun and then of Mars, produces earthquakes, and so does the nebulous one on the eye of the Archer. What is of the mixture of Mars, [like] al-Dabarān, which is of the greatest magnitude, is fiery and generates lightning and thunderbolts. [Stars] like the Heart of the Lion, whose mixture is first that of Mars and then of Jupiter, produce stagnant hot air and pestilence. Those of them of the mixture of Venus alone signify much moisture and rain; those of the mixture of Jupiter, moderate heat and moisture; those of the mixture of Saturn alone, cold; and those of the mixture of Mercury alone now dry and now moisten, and generate winds, according to what we have mentioned of the powers of the wandering stars and of its familiarity with the winds. After that their natures are mixed with the natures of the signs in which they are and with the natures of the seasons of the year. The

[36] Ptolemy (*Tetrabiblos* II.11) also divides each sign into a leading, a middle and a following part, which al-Battānī leaves out. For the northern part of Libra Ptolemy has «windy», and for the southern part of Aquarius «cloudy»; for Cancer the transcription has للصحر, read للصحو, «fair weather».

moist stars in the western quarter [are moister], and likewise the hot stars in summer increase in heat through their familiarity with the season, and when they are in the southern quarter their heat is greatest. When the signs in which they are suit their natures, that increases their action more; likewise, when they incline in latitude to the north or to the south, that increases or decreases their action, according to how far their natures suit that direction. When the mixture of the stars is attributed to two of the wandering stars, the first of the two is stronger in action, and the second mixes with it afterwards, according to the marks, for some of them resemble the second star only a little in their mixture; [and it depends] on how the wandering stars connect with one another when they take over the governing, in the familiarity of their natures with one another and in their contrariety; so that from the whole of what comes together from these mixtures you arrive at the condition that will be, and give the preponderance in power to what is most manifest in action and strongest in position.[37]

Since we have first set out these things in this way, the methods of the particular indications are as I shall describe. One of these methods is the consideration of the matter as a whole, which is known from considering the condition of each of the quarters [of the year]. In it one needs to consider what we have said: to examine the conjunction of the sun and the moon and their opposition that come nearest before the points of the solstices and the equinoxes, and to set up the cardines, at the time of the conjunction or the opposition, for each of the climes and the cities that one intends to examine. Then after that take the planets that govern the places in which the conjunction or the opposition takes place, and those that govern the cardine that follows these places, that is, the cardine through which one of the luminaries passes shortly before the time of the conjunction or the full moon, as we have explained and set out in the chapter on eclipses; so that the governor among the wandering and the non-wandering stars is known in that way, and we know the matter as a whole from the properties of the quarters, and know the increase and decrease in the severity or mildness of that whole from the nature of the governing planets, by looking at their quality, which of the qualities it is, and what conditions are set in motion by them.

f. 33r

The second method is the consideration of the affairs of the months. In it one needs to look at the conjunctions and oppositions that occur in each of the signs in the very way mentioned, keeping in mind what I shall mention: if the conjunction falls near the point of the solstice or of the equinox that has been passed, we use the conjunctions until [the sun] reaches the quarter that follows that quarter. Likewise one should look at the cardines and at the planets that govern both the places that we have mentioned, and examine in particular the nearest appearances of the planets, the applications and separations of the wandering stars, their properties and conditions, and which winds these planets and the parts of the signs in which they happen to be stir up. Also, as we said, we look towards the quarter of which wind the latitude of the moon inclines when it inclines from the belt of the signs; so that, proceeding [in this way], one knows from all these things what for the most part prevails in the conditions of the months and their winds.

f. 33v

The third method is to examine the finest indications, which show the slackening and the severity of the matter. This is known from the sharing of the sun and the moon in particular configurations, not only in the configurations of conjunction and opposition but also in those at the half of the moon in light; and the beginning of the change indicated is for the most part three days before the moon's progress [matches that of] the sun, and sometimes three days after, at equal distances. It is also known from their sharing in configurations with the wandering stars in each of these conditions and in others, such as the condition of the trine and the sextile; for these things follow the natural course. The condition of the change we come to know by following

[37] «Nebulous» reads السحابيّة for السماويّة, «heavenly», and «the Manger» المعلف for العلف; the words «of the fixed stars in each sign» stand twice, and the text has «Mars» before «the nebulous one on the eye of the Archer» and «the Head» (الرأس) for «the Archer» (الرامي). Ptolemy (*Tetrabiblos* I.9) gives the cluster after the sting of the Scorpion the mixture of Mars and the moon, and the cluster in the face of the Archer that of the sun and Mars.

the natural familiarities that the stars that are present, and the signs, have with the surrounding air and with the winds. The strength of these particular conditions that occur from day to day is [greatest] especially when the fixed stars that are greatest in size and strongest in action make appearances in the mornings or in the evenings, in the east or in the west, relative to the sun; for for the most part they change and turn the particular conditions to their own natures. This also happens, a little less than what we have said, when the luminaries pass one of the cardines; for the slackening and severity that the conditions have from time to time occur when the luminaries come to these places, just as the ebb and the flood occur when the moon comes to them; and the change of the winds in particular occurs when the luminaries go to the cardines, the change being towards the wind to which the latitude of the moon inclines. In every case one should proceed on the basis that the first, universal cause takes precedence, that what follows it is the cause of the particular accidents, and that the action is most certain and strongest when the stars that govern the universal natures are found sharing in a configuration with the particular natures.[38]

## ⟨14⟩ On the indications taken from the phenomena of the upper air[39]

f. 34r

In foreknowing the particular indications it is useful to examine the signs that appear around the sun, the moon and the stars at their times. One should examine what appears of this in the sun: when it rises, in seeking the conditions of the day; when it sets, in seeking to know the conditions of the night; and, in knowing the conditions of longer duration, at the times when the sun shares in a configuration with the moon, since each configuration for the most part signifies the condition from that time to the time when the moon reaches the configuration that follows it. When the sun rises or sets clear, without anything covering it, pure and without cloud, it signifies fair weather. If its disc is of varied colours, or inclines to fiery red, or the rays that go out from it and that are around it are red, and the clouds that are called «suns» are on one side of it, of the colour of cloud inclining to red, and the rays of the sun are stretched out long, it signifies strong winds, and these winds blow towards the angles in which the signs mentioned are. Instead of the words «the clouds that are called suns» one may also say «what is beside the sun». If it rises black or inclining to green, or sets with clouds, or has around it one halo or two, or the clouds called suns are on both its sides, and its rays incline to green or are black, it signifies wintry weather and rain.[40]

As for the moon, its position should be examined on the day three days before the conjunction, three days before the opposition and three days before the half in light, [and] three days after these times. If it appears thin and clear then, with nothing around it, it signifies fair weather. If it is thin and red, and all of its circle that has no light seems to be in motion, it signifies the blowing of winds from the regions towards which it inclines. If it is black or green and thick, it signifies wintry weather and rains. One should also examine the halo around it. If there is one, clear and fading little by little, it signifies clear weather; if there are two or three, wintry weather. If they incline to bright red and seem broken, they signify wintry weather caused by strong winds; if they are turbid and thick, wintry weather caused by snowstorms; if they incline to green, black and broken, wintry weather of both kinds; and the more haloes there are, the more the things we have mentioned increase.[41]

f. 34v

Again, the halo around the wandering stars and around the bright stars that are not wandering

[38] «The change» reads التغيير for التفسير, «the interpretation», and «the half [of the moon]» انتصاف for اتصاف.

[39] *Tetrabiblos* II.13.

[40] The transcription adds after «with the moon» the words «not when it rises or sets clear, without anything covering it», which recur below. The «suns» are Ptolemy's mock suns, the parhelia. «Halo» follows the Arabic *Tetrabiblos* (هالة) for the transcription's حالة here and below.

[41] «Thin» follows the Arabic *Tetrabiblos* (دقيقًا), and «snowstorms» likewise (الدمق); the transcription has فيها and الزمن.

signifies things [akin] to the colours of those haloes and to the natures of what they surround. One should also examine the stars that are not wandering and look at the groups of them in which there are many stars close to one another, and examine their colours and sizes; for when they appear brighter and larger than is usual with them, it signifies the blowing of winds from the region in which the stars that appear are, whatever region of the sphere it may be. Likewise, when the stars that resemble smudges, especially like the star called the Manger, I mean the one in the figure of Cancer, and the like, are hidden in fair weather and do not appear; or the things that occur at some times among the phenomena of the upper air: for the smudge-like stars with tails always signify dryness of the air [and] winds, the strength of that depending on the number of the things and on the length of their stay.[42]

As for the falling of stars and their running: if it is from one angle, it signifies the blowing of a wind from that region; if they fall from different angles, they signify winds that blow without order or arrangement; and if it is from the four angles, they signify wintry weather of many changes, going as far as the occurrence of thunder, lightning and the like. So too the clouds: when they resemble tufts of wool, they sometimes signify wintry weather. And the rainbow that appears at some times: when it appears in fair weather it signifies wintry weather, and when it appears in wintry weather it signifies fair weather. I say, in general, that for the most part the particular colours that appear in the air signify conditions that suit the accidents that the colours mentioned before signified.[43]

## ⟨15⟩ On knowing the matters of nativities and what is inferred in them[44]

The first thing we should consider among the matters of nativities: since people have a temporal beginning of their coming to be, and this beginning belongs to this very kind of consideration, for whoever wishes to look closely and to delve into the examination of the properties that are in the composition itself alone.[45]

## ⟨16⟩ On knowing the degree of the ascendant of the native[46]

f. 35r

Since doubt often arises about the first matter on which one relies, the degree of the ascendant at the time of the coming out of the child — for for the most part only those who observe it with the astrolabe, alone among all the instruments for the hours, know the hours at the time of birth accurately, while the other instruments for the hours, with which many who are concerned with this measure, may often be false and depart from the truth: those of them by which the hours are measured by the sun, because of the inclination that may befall them in their place and the crookedness of their gnomons, and the water clocks, because of the stopping of the water and the inequality of its flow, which come from various causes and from the least cause that befalls it — it is necessary first to report the way by which one can know the degree of the sphere of the signs that, by the natural, necessary reasoning, must be the one rising, after we have found approximately, from the hour that we have been given, the degree of the ascendant from the ascensions. We say that one should take, of the conjunction and the opposition that came before the birth, the one nearest to it, whichever was nearer, and find its degree accurately. If what

[42] «The Manger» follows the Arabic *Tetrabiblos* (المعلف); the transcription has القلب, «the Heart». The words «likewise» stand twice. The sentence on the clusters lacks its conclusion; Ptolemy says that when they are dim or invisible in a clear sky they signify heavy rains.

[43] «Falling» follows the Arabic *Tetrabiblos* (انقضاض) for the transcription's انفضاض, «dispersing»; the copyist repeated «and the rainbow that appears» and struck the repetition out.

[44] *Tetrabiblos* III.1.

[45] The sentence is incomplete. Ptolemy's introduction to the nativities (*Tetrabiblos* III.1) takes as the starting point of a human being the time of conception or, failing it, of birth.

[46] *Tetrabiblos* III.2.

we have taken is a conjunction, we take the degree of the luminaries; if what was taken is an opposition, we take the degree of whichever of the luminaries happens to be above the earth.[47]

f. 35v

Then we look which of the planets deserves to have the governing of that degree at the time of the birth. In general, the way in which a planet deserves the governing is completed by five things, I mean the triplicity, the house, the exaltation, the term, and the appearance near the rays or the sharing in a configuration with the sun; so the governing of the place belongs to the planet that has more than one of these qualifications in that place, or all of them. If we find all or most of these things in one planet, we look at its degree, taken accurately, in the sign in which it is at the time of the birth, and judge that the degree of equal number in the sign that we had taken approximately as the ascendant is the degree of one of the cardines of the ascendant. If we find that two planets or more share in the governing, we look at the number [of degrees] that each of them has travelled at the time of the birth; the one whose number is nearest to the number of the degree of the ascendant that came out for us from the ascendant [found approximately] is the one that rises at the time of the birth. If we find the number of the degrees of two planets or more near the number of the degree of the ascendant that came out for us, we follow in that the planet that has more shares in the cardines and in whose domain [it is]. If the distance of the degree of the governing planet from the degrees of the ascendant taken roughly is greater than its distance from the degrees of the middle of the sky, [we take it for the middle of the sky] and set up the other cardines from it.[48]

One should look at the latitude of the planet, since it does not stay on the belt of the signs, so that the degree of the sphere of the signs with which the planet rises, and the degree with which it culminates, are known, because of the latitude, as we have explained at the beginning of the book. Then we find the degree of the ascendant and the degrees of the middle of the sky that we found approximately from the ascensions. If the degrees of the planet are nearer in number to the degrees of the middle of the sky than to the degrees of the ascendant, we make the number of the degrees of the middle of the sky equal to the degrees with which the planet culminates. Disagreement may arise in what is taken with the astrolabe in various ways, and for that reason this chapter has become a standard for all.

## ⟨17⟩ On the division of the doctrine of nativities[49]

f. 36r

Having first set these things down, we say: anyone who wishes to divide the whole consideration of nativities in an orderly and systematic way finds that some of the things that can be grasped by the natural course are the accidents that occur before the birth, like the account of the parents; and some are the accidents that occur before the birth and after it, like the account of the brothers. Those at the birth itself are not one simple matter, like what we have described. The last of the accidents that can be grasped are those that occur after the birth, and this kind has many branches and parts. What one wishes to know of what is at the time of the birth itself is the account of male and female, the account of those born as twins and as more than twins, the account of the malformed, and the account of those who are not nourished. What one wishes to know of what comes after the birth is the account of the length of life, since it is not connected with the account of those who are not nourished; and of that is the account of the form of the body and of the illnesses and injuries that befall the body. After that comes the account of the conditions of the soul and of the accidents of the soul; after that, the account of the native's fortune in the matter of possessions and in the matter of rank and elevation; after that, the account of the native's condition in the works they engage in; after that, the account of their marriage and married life,

[47] «Doubt» reads الشكّ for الشدّ. «The astrolabe» is ذات الصفائح, as in the *Zīj*; the Arabic *Tetrabiblos* has الأسطرلاب. The copyist wrote «the lord of the sphere» before «from the ascensions» and struck it out.

[48] «Term» reads الحدّ for الجدّ.

[49] *Tetrabiblos* III.3.

of the children born to them, of their mixing with people and their partnerships with them, and of their friends; after that, the account of travel abroad; and at the end, the account of the condition of their death, which in its power is akin to the account of the length of life, and whose place must be after all this. We shall describe the matter of each of these things in outline.[50]

Since we have explained the degree of the ascendant and its correction in [this] way, we add [an account] of everything that occurs in the conditions of nativities, and mention it together with our mention of the pure active powers, as we said just now. We discard what many people have described in which there is no proof or convincing argument suited to the causes that come from the first nature, and examine what of it can be grasped, not by lots and numbers that have no cause that necessitates them, but by considering the configurations of the stars and their fitness to their places. So that there is no need to repeat and reiterate the account in each of the chapters that we have mentioned, we should give a general account for all of them.[51]

f. 36v

We look first at the place in the sphere of the signs that suits the kind of the chapter sought among the chapters of the consideration of the nativity: as the place of the middle of the sky, for example, suits the matter of the works one engages in, I mean the crafts, or as the place of the sun suits the matter of the father. Then after that we look at the planets that deserve the governing of that place by the five kinds that we mentioned in what we said before. If the lord of the place is one planet in all respects, we give it the governing of that matter; if there are two planets or three, we give the governing to the one of them that has the greater number of shares. After that we direct our consideration of the condition of the event, and our examination of it, to the natures of the very planets that have deserved the governing, and to the natures of the signs in which the planets are, and of the places akin to them. Then we examine the event and its size from their strength: we look whether they are strong in their condition in the world, or in their condition in the nativity, or the matter is the opposite. They are strongest in their condition in the world when they are in places of their own or akin to them, and when they are eastern and increasing in motion; strongest in their condition in the nativity when they are in the cardines, or in the places that rise to the cardines, and especially the ascendant and the middle of the sky; weakest in their condition in the world when they are in foreign places, or are western, or retrograde, or decreasing in motion; and weakest in their condition in the nativities when they decline from the cardines. Direct your consideration of the length of the time of the event by examining whether they are attributed to the mornings or to the evenings, relative to the sun or to the ascendant: for the two quarters that precede each of these, and the two opposite them on the diameter, are attributed to the mornings, and the remaining quarters that follow them are attributed to the evenings. And look whether they are in the cardines [or] rising to the cardines: if they are attributed to the mornings and are on the cardines, they are quicker in action; and if they are attributed to the evenings and rising to the cardines, they are slower in action.[52]

One should also keep in mind, in all the matters that arise from nativities, that a planet that brings about something in the place sought must necessarily have been akin to it from the beginning. For if it is not akin to it, it cannot bring about anything in it, since it did not share in the first beginning: that is, if the sign in which the planet is is not joined to the sign sought into which the planet falls, it cannot bring about anything of its nature in that sign, since it was not joined to it in the nativity. The time when the event comes about does not follow the first governing, but follows the condition of the planet that has dominion over the governing, relative to the sun and to the angles of the world.[53]

[50] «The account of the malformed» is written in the margin. «We shall describe» follows the Berlin manuscript (سنصف); the transcription has ويتّصف.

[51] «Lots» reads القرع for الفرع: Ptolemy rejects the method of lots and numbers (*Tetrabiblos* III.3).

[52] «Matter» follows the Arabic *Tetrabiblos* (لأمر) twice for the transcription's لأمن, «safety»; «their strength» and «their condition» read قوّتها and حالها for قوّتنا and حالنا, «our strength», «our condition».

[53] «If it is not» supplies the negation (لم) that the Berlin manuscript has.

## ⟨18⟩ On the parents[54]

f. 37r

The sun and Saturn are by nature akin to the fathers, and the moon and Venus akin [to the mothers]; and as the condition of these planets is found to be with respect to one another and to the others, so we must suppose the condition of the parents will be. For one should consider their fortune (?) and their possessions from the attendance of the planets on the luminaries. When the beneficent planets of their own domain surround them, those planets being either in the signs of the luminaries themselves or in the signs that follow them, they signify that the affairs of the parents will be splendid and noble; especially when the planets attending the sun are attributed to the mornings and those attending the moon to the evenings, that is, when they rise before the sun and after the moon, so that their position is praiseworthy for that reason. If, besides, Saturn and Venus are eastern, or in the facing proper to them, or on the cardines, we should suppose from that a manifest good fortune of the parents, according to what suits each of them. If the matter is otherwise, and the luminaries proceed alone with no planet attending them, they signify the low station and meanness of the parents, especially when the position of Venus and Saturn is not praiseworthy. If they are attended by maleficent planets not of their domain, as happens when Mars rises after the sun and Saturn after the moon, and the position of the beneficent planets relative to the luminaries is not praiseworthy and they are not of their domain, one should infer from that the meanness of the parents' living and the unsettledness of their condition. If the lot that signifies fortune, which is the lot of good fortune that we shall mention, agrees in the nativity, in a praiseworthy agreement, with the planets attending the sun or the moon, the possessions of the parents remain safe. If the lot is not as we have said and the matter is otherwise, and then none of the beneficent planets attends the luminaries, or those that attend them are maleficent, there is no benefit from the possessions of the parents, and they become harmful to them.[55]

f. 37v

The length and the shortness of the parents' life are considered from the other sharings of the planets in configurations. As for the father's life: if Jupiter and Venus share in a configuration with the sun or with Saturn, in whatever way, or if Saturn too shares with the sun, its sharing with the sun being one of agreement — either being with it, or in its trine, or in its sextile — and both are strong, it signifies long life for the father. If they are not strong, it is not so; but they do not necessarily signify short life. If the matter is not so, and Mars is above the sun or above Saturn, or follows them — that is, it rises after them — and Saturn is not in agreement with the sun but in its quadrature or in opposition to it on the diameter, or both are declining from the cardines, weakness of the fathers comes from them. If they are on the cardines, or where they rise to the cardines, they make the fathers short-lived or bring an injury upon them: shortness of life they bring about if they are on the ascendant or on the middle of the sky, and injury if they are on the two remaining cardines, I mean the setting point and the cardine of the earth, or where they rise. For when Mars looks at the sun in the way that I mentioned at the beginning of the book — that is, the parts that we said look at one another and are equal in power — the father dies suddenly, or an injury to the eyes befalls the father. If Mars's look is towards Saturn, there befalls the father either death, or fever with shivering, or an injury from cutting or burning. Saturn too, when its configuration with the sun is bad, brings about the death of the fathers by diseases and by illnesses that befall them from harmful humours.[56]

f. 38r

As for the mother's life: if Jupiter shares in a configuration with the moon or Venus, in whatever way, or Venus is in agreement with the moon, either being with it, or in its sextile or its trine, and both are strong, they signify long life for the mother. As for Mars, if it looks at the moon, or follows Venus, or is in quadrature or in opposition to it, or Saturn is in this condition with

[54] *Tetrabiblos* III.4.

[55] «Will be» is written between the lines. «Suits» follows the Arabic *Tetrabiblos* (يلائم).

[56] Robbins takes Ptolemy's «in the way described» as quadrature, opposition or overcoming; al-Battānī refers it to the signs that look at one another and are of equal power (*Tetrabiblos* I.15; chapter 6 above).

the moon: if they are decreasing in motion, retrograde, or declining from the cardines, they bring upon the mothers only a falling into illnesses and sicknesses; and if they are increasing in motion or on the cardines, they shorten the mothers' lives and bring [injury] upon them. They shorten life when they are on the two eastern cardines or where they rise to them, and bring about injury when they are on the two western cardines or where they come to them. For when Mars looks at the moon in the way we have mentioned, and the moon is eastern, sudden death of the mothers comes of it, or an injury that befalls them in the eyes; if the moon is western, their death comes of it from miscarriage or the like, or an injury from cutting or burning. Instead of the word «western» one may also say «declining». If Mars looks at Venus, their death by fevers comes of it, and illnesses that befall them from darkness of sight and from matter that flows down suddenly. When Saturn looks at the moon, death and illnesses befall them: if the moon is eastern, their death is from fevers with shivering; if it is western, it is from spreading tumours and ulcers in the womb.[57]

f. 38v

In knowing the kinds of the particular injuries and illnesses, or the death that befalls the parents, we should use, besides what we have mentioned, the conditions of the signs in which the planets that bring about the particular condition are found; we shall mention the conditions of the signs and explain them when we speak of the native. And we should examine by day in particular the sun and Venus, and by night in particular Saturn and the moon. What befits us, and is required of us, in examining the remaining particular matters of the parents is to set up the place of the father or of the mother of the [proper] domain in the place of the ascendant, and to consider all their other matters as is done in the nativities of the parents themselves, according to the methods, accidental and active, that are found in the general kinds, which we shall explain later. The ways in which the causes combine with one another we must remember here and in all the places that come later in the matters of nativities: if the planets that are the lords of the places sought are not of one kind but of different kinds, producing opposites, we look which of the planets that are strongest in power, dominion and prevalence we find to have the most shares, so that it has the best claim to what happens; and in examining them we always keep to the natures of these. If the shares are of equal weight, then if the planets are [together] (?), we arrive, with what we have described, at knowledge of what comes together from the mixture of these different natures with ease; and if they are apart, we give each of them the accidents that suit it according to the particular times, giving first to those of them attributed to the evenings.[58]

f. 39r

Others than Ptolemy have said the like about the indications of the parents, except that they added to it, by night and by day, the lot of the father and of the mother and the sign of the parents, and considered the strength and weakness of all the significators, their positions relative to the cardines, and the lords of these places. If they found them strong, they inferred the excellence of the parents and their good condition; if they found them fallen and weak, they inferred their meanness. In this they also used the lords of the triplicities of the significators, and said: if, with the corruption of the significators, these are afflicted and corrupted with them, they signify chronic illness of the parents. If a beneficent planet is found in the house of the parents, and the lords of the houses and of the terms, and the terms of the luminaries, look at it, they judged the parents to be noble; if the maleficent planets look at the house of the parents from quadrature or opposition, or are in it, and the beneficent do not look at it, they judged that they are slaves or like slaves. If the sign of the parents is the house of the sun, the indication is for the father; if it is in the house of the moon, the indication is for the mother. If you find the lord of the twelfth

[57] For «eastern» and «western» of the moon Ptolemy has «waxing» and «waning». «Shivering» reads نافض for ناقص.

[58] «Of the domain» reads للحيّز for للخير, «of the good»: Ptolemy has «the paternal or maternal place of the sect» (*Tetrabiblos* III.4). «Causes», «lords» and «dominion» follow the Arabic *Tetrabiblos* (الأسباب، أرباب، استيلاء) for الأشياء، أرباع، استعلاء; «we arrive» likewise (وصلنا). «Together» is a guess for the transcription's «the high» (العالية): Ptolemy has «when the rulers are together», and gives the events first to the more eastern planets, not to those attributed to the evenings.

house with the moon in the thirteenth (?) from the cardine, the native's mother is a slave-woman; if this is the condition of the sun, the native's father is a slave. If the lords of the triplicities of someone's significator are in their own dignities, in praiseworthy places of the sphere, free from the maleficent planets, with the beneficent looking at them, they signify gain and rank in what is to come; especially when the lot of good fortune and its lord are fortunate together with the lot of the parents, for that signifies the continuance of the matter and its inheritance by the sons. If the matter is the opposite, the parents have no benefit from their possessions, and [the significators] signify the fall of their rank.[59]

They considered the agreement of the parents with the native and their love for the native. One looks at the planet that prevails over the ascendant and the one that prevails over the matter of the father by the number of shares: if they agree and suit each other, they love one another; if the matter is the opposite, they hate one another. Likewise, if the sun is in the term of Mars by day, or of Saturn by night, especially when the maleficent planets look at the sun and the beneficent do not, it signifies mutual hatred, or the native is deprived (?) of rank and of wealth. If the moon is as I have described of the condition of the sun, the mother meets with affliction from the child. [The parent] whose lot is found in the twelfth [place] from it, the native will be an enemy to that parent. When the sun is in a cardine and Mars looks at it from opposition, the father hates the child; likewise, when the moon is in this condition, the mother hates the child. If Jupiter and Saturn are in opposition and overlook one another, it signifies mutual hatred, and if they are in quadrature likewise; from the sextile and the trine it signifies love. If the lord of the house of the parents does not look at the parents, nor the lord of the sun at it, nor the lord of the lot of the parents at the lot, the native [will have] sorrow (?) and hardship. If the sun is afflicted by day from opposition and quadrature and is in a bad place, the father hates the child and drives the child away (?).[60]

f. 39v

They said about the life of the parents: if the sun looks at the ascendant by day, it is directed for the life of the father; if Saturn looks at the ascendant, the direction is from it, if the sun is fallen; if Saturn is fallen, the degree of the middle of the sky by day, and by night the degree of the cardine of the earth; and, by night and by day, the lot of the parents. Compare what comes out of the direction with the years of the planet that has the most shares in the place that you direct for the father: if it agrees with them, or is near their number, it signifies the death of the father. Use the like for the significators of the mother. Some of them looked by day and began with the sun for the father: if it is joined to the ascendant, with one of the lords of the shares that follow its place looking at it, they used it for the life of the father; if one did not find it so, one looked at Saturn; if one did not find it in this condition, one looked at the middle of the cardine of the sky, provided that the lot of the father does not have this condition. By night you begin with Saturn, then the sun, then the lot of the father, then the cardine of the earth; and you make the planet that has a share and a look a helper of the significator of the moon (?), according to its years. Use the like for the significators of the mothers.[61]

### ⟨19⟩ On the brothers[62]

f. 40r

As for the matter of the brothers: anyone who examines it in general, without going beyond the possible measure in seeking the number of the brothers accurately, one by one, can know the matter of the brothers who are of one mother, by the natural method, from the sign that is in the middle of the sky from the place of the mother, I mean from the place of Venus by day and the place of the moon by night among the signs. Since this sign, and the sign that rises to its place, are the place from which the children of the mother are known, that must be the place

[59] «Significators» reads الأدلّاء for الأولاد, «the children», and «fortunate» مسعودين for سعودين منحوس.

[60] «Sorrow» guesses at the transcription's {سعود}; the Berlin manuscript seems to have لغمر.

[61] «The death» reads هلاك for هلال, «the new moon».

[62] *Tetrabiblos* III.5.

of the brothers. If the beneficent planets share in a configuration with these places, we judge from that that the brothers are many, base our conjecture on the number of these planets, and look whether they are in signs of one form or in bicorporeal signs. If the maleficent planets overpower the place, or face it in opposition on the diameter, being in the house opposite it, they signify few brothers, especially when the sun enters with them. If the sun (?) is on the cardines, and especially on the ascendant, and then the planet in one of these places is Saturn, it is an indication of this among the first-born; if it is Mars, it lessens the number of the brothers by death. As for the planets that give brothers: if their condition, in their position in the world, is praiseworthy, we should suppose that the brothers will be persons of standing and noble; if the matter is the opposite, they will be lowly and obscure. If the maleficent planets overpower the giving ones, or follow them, they will be short-lived too. Those of them that give males are the planets masculine in their condition in the world, and those that give females the feminine in their condition in the world; again, the eastern ones give the first brothers and the western the last. Besides this, if the giving planets share in an agreeing configuration with the sign in charge of the matter of the brothers, they signify that the brothers love one another; if they agree with the lot of good fortune, they signify that the brothers share in their affairs; and if they are in signs between which there is no connection, or in the condition opposite to the one we have mentioned, they signify that the brothers are hostile and envious of one another and, for the most part, cunning. Anyone who wishes to examine each of the matters of the brothers more than this can also conjecture about it by setting up the giving planet in the place of the ascendant and considering all their other matters as one considers a nativity.[63]

f. 40v

Others than Ptolemy among the ancients looked into the matters of the brothers from the third house from the ascendant, Mars, the lot of the brothers, and their lords. If most of these significators, especially the planets that have many shares in these places, are in the signs of many offspring, and the sign of the house of the brothers likewise, they signify many brothers; if they are otherwise, they signify few. If most of them, especially those with many shares, are in the masculine signs, most of them are males; if in feminine signs, most of them are females. Speak of their safety and death, and of the soundness and corruption of their condition, from the soundness of the condition of the significators and of the house of the brothers, and from their corruption. Look at the powers of the lord (?) of the triplicity of the ascendant. If it is in the ascendant, [the native] is a first-born; if it is in the middle of the sky, a first-born or a fourth; if in the seventh, a first-born or a seventh; if it is in a place other than a cardine, above the earth, count from it to the ascendant: if there is a maleficent planet between them, it signifies that some of the children miscarried or died; if they are beneficent, or in some of their own dignities, strong, in agreeing signs and in praiseworthy places, they signify the number of those born before the native. If the lord of the triplicity is below the earth, count from the ascendant to it, as we said first. They also looked, for the number of those born before the native, at the number of the planets found between the middle of the sky and the ascendant: if no planet is found there, the native is a first-born, or if one was born before, it died; and, for the number of those born after the native, at the number of the planets found between the ascendant and the fourth. They said that when the moon and the ascendant of the native are in signs of many offspring, the mother has many children. If the moon is found in Scorpio, in three degrees of it, the mother has no children, or if they were born they did not survive but died. If we find both lords of the triplicity of the ascendant in the places third from the cardines, the native has no brothers, or if there are any, they are scattered and do not come together in the land. If you find the lord of the house of the brothers in the ascendant, the native is single, with no brother; if it is found in the middle of the sky, the native has an older brother; and if it is found in [the house of] the parents, a younger brother.[64]

f. 41r

[63] «Of the mother» follows the Arabic *Tetrabiblos* (الأمّ) for الأمر; «conjecture» follows the Berlin manuscript (يحدس). Where the transcription has «the sun», Ptolemy has «if the opposition is at the angles».

[64] «Most» reads جلّ for حلّ, and «the significators» الأدلّاء for الأولاد, «the children», as in chapter 18. «The lord»

If the house of the brothers is found to be a bicorporeal sign, and its lord also in a bicorporeal sign, especially when the sign in which the lot of the brothers is is so too, it signifies that the native will have brothers from another father or from another mother. For their love and enmity one should look at the lot of the brothers and its lord: if they look at one another from trine or sextile, it signifies mutual love; if from quadrature, a middling friendship; if from opposition, enmity; and if they do not look at one another, it signifies estrangement and seldom meeting. The same is known from the look of the planet that prevails over the ascendant by its shares at the one that prevails over the house of the brothers, the lot of the brothers and Mars. If the lord of the lot of good fortune is with the lot of the brothers, and the lord of the lot of the brothers with the lord of the lot of good fortune, it signifies that they benefit one another, and they may get much wealth from their brothers. For the death of the brothers one looks to the lot and to the lord of the lot of the brothers, at the conjunction, quadrature and opposition of the maleficent planets, especially when the maleficent planet is in its first station or retrograde, and above all when Jupiter does not look at the lot. If one of the two maleficent planets looks at the lot while it is in the dignities of Mercury, it signifies that the native's brothers are killed; and if one of the two maleficent planets is in the sign in which Mercury is and looks at the lot of the brothers from opposition and quadrature, we judge likewise. For the sisters Venus is used in the place of Mercury.

### ⟨20⟩ On males and females[65]

Knowledge of the male and the female sex of those born comes from both luminaries, from the ascendant and from the planets that have shares in them. It is known especially from the condition at the time of the falling of the seed, and in general from the condition at the time of the birth too. One should examine these places that we have mentioned and the planets that govern them: if they are all, or most of them, masculine, the child is male; if feminine, female. One should distinguish the masculine and the feminine places and planets in the manner that we mentioned at the beginning of this book: I mean from the nature of the signs in which they are, from the nature of the planets themselves, and from their condition relative to the world — for they become masculine when they are eastern of the horizon and feminine when they are western — and also from their condition relative to the sun, for they become masculine when they are attributed to the mornings and feminine when attributed to the evenings; and from all of that one conjectures which sex will for the most part prevail in the child.[66]

### ⟨21⟩ On the birth of twins[67]

f. 41v

Likewise, in the birth of twins or of more than twins, we should examine the matter of these three places, I mean the luminaries and the ascendant. It is in the nature of this matter to occur at the minglings when two of these places, or all three of them, are bicorporeal signs, especially when this same condition also befalls the planets that govern them; or when some of them are in bicorporeal signs, or some are [disposed] two by two or more than two in the governing. If the chief places are bicorporeal signs and share in a configuration, and with that many of the planets [are so too], those born will be more than two, and the planet that has this property increases their number. The kind of the planets that share in a configuration with the sun, the moon and the ascendant, which increases the generation of males or females, as mentioned before, adds to the knowledge of which of them are male and which female. If this condition does not include the sign of the ascendant together with the luminaries, but the sign of the middle of the sky is

guesses at the transcription's {زلي}.

[65] *Tetrabiblos* III.6.

[66] «Conjectures» reads فيحدس for فيحدث.

[67] *Tetrabiblos* III.7.

found in this condition instead, what the mothers bear is for the most part twins or more than that.[68]

In particular, for the birth of three males, complete and well formed, the matter is like that of the birth of the Anaktores, who are the kings: Saturn, Jupiter and Mars in the places we have mentioned, in bicorporeal signs. For the birth of three females, the matter is like that in the birth of the Charites: Venus and the moon with Mercury, made feminine, in the places we have mentioned. For the birth of two males and a female, the matter is like that in the birth of those called the Dioscuri: Saturn, Jupiter and Venus in these places. For [the birth of] two females and one male, the matter is like that in the birth of the one called Demeter and Kore: Venus, the moon and Mars in these places. In these conditions it mostly happens that the child is not complete, that its birth is with some injuries that befall the body, and that things arise in some of its limbs without visible cause, of an unexpected kind; or some of these accidents are on it in the manner of something that appears [as a portent].[69]

## ⟨22⟩ On the malformed[70]

f. 42r

Not outside this subject is the account of the malformed. We say first that in such cases the luminaries are for the most part found declining, or not joined to the ascendant, and the maleficent planets occupy the cardines. When we find the matter in this condition — since it is something that often happens in lowly nativities of ill fortune, even if those nativities are not of the malformed — we should look at the preceding conjunction and opposition, and find the planet that governs the place of either of them, and those that govern the place of the luminaries at the time of the birth. If their places in the nativity, and the place of the moon or the place of the ascendant, all or most of them, are not joined to the place of the conjunction or the opposition, the child will be of changed form. If these things are so, and we find the luminaries besides in the four-footed signs or in signs of wild beasts, and the planets on the cardines are maleficent, the child will not be of a human being.[71]

f. 42v

If none of the beneficent planets witnesses to the luminaries, and the maleficent witness to them, the child will be untamed, and its nature one of the harmful wild natures. If either Jupiter or Venus witnesses to the luminaries, the child will be of the animals that are familiar with people, like dogs, ichneumons and the like. If Mercury witnesses, the child will be of the animals that people need and use, like pigs, fowls, cattle, goats and the like. If the luminaries are in signs whose figures have the form of human beings, and all the other things are likewise, the child will be of ⟨human beings⟩ or of what is near to human beings, except that it will be malformed and changed. The knowledge of its condition, here especially too, comes from the forms of the signs in which are the maleficent planets that surround the luminaries or the cardines. If here too none of the beneficent planets witnesses to any of the places we have mentioned, the child will be without speech, and its form truly and completely changed. If Jupiter or Venus witnesses, the property of the child that has the defect is that it is honoured and respected: for from this arises the form called Hermaphroditos, which means the form attributed to Mercury and Venus, and the form called Harpocratiac, and the like. If Mercury witnesses too, the child, with these things, will be an interpreter of dreams, and their earnings and resourcefulness will come from these things; but they will be toothless and deaf, though in other things of good nature, a person

[68] «Minglings» follows the Arabic *Tetrabiblos* (شيئين is its reading of the next word); Ptolemy speaks of the intermixture at conception.

[69] The Greek names, corrupt in the transcription (نطروس، الحارطين، دياسفودا، لمنظر وفودس), are read from the Arabic *Tetrabiblos* (أناقطرس، الحارطس، ديوسقور، ديميطر وقوريس): Ptolemy's Kings, Graces, Dioscuri, and Demeter and Kore.

[70] *Tetrabiblos* III.8.

[71] «Something», «governs» and «those that govern» follow the Arabic *Tetrabiblos* and the Berlin manuscript (شيئًا، المدبّر، مدبّري).

of learning and of cunning.[72]

## ⟨23⟩ On those who are not nourished and those who are not reared[73]

f. 43r

Since there remains, of what one wishes to know of what is at the time of birth, the account of those who are not nourished, one should know that this account is in some respects joined to the account of the length of life, since it is not far from each of these two kinds, and in some respects distinct from it, in that the force of this inquiry differs somewhat. For the account of the length of life concerns all who have a perceptible life, those whose life is not shorter than one of the revolutions of the sun, I mean a year; and life in potency is also what is less than this, months, days and hours. The account of those who are not nourished is the account of those who reach none of the times we have mentioned, because of the excess of evil and harm upon them; and for this reason the inquiry into the length of life has many parts, and the inquiry into those who are not nourished is general and simple.[74]

f. 43v

In general, when one of the luminaries is on one of the cardines, and [one of] the two maleficent planets shares with it in longitude, either being with it degree for degree, or in opposition, or in an isosceles configuration, and none of the beneficent planets shares in a configuration with it, and the governor of the place of the luminaries is found with the maleficent planets, the child is not nourished and dies at once. If it is not in an isosceles configuration, but the casting of the rays of the maleficent planets is near the place of the luminaries, rising to them — that is, the luminaries are near them, in the parts that come after the parts of the luminaries in the signs in which they are — or if the maleficent planets are two and harm one of the luminaries or both: either by rising to their place, or to the place of one of them, as we have said, by conjunction; or by facing it, or both of them, on the diameter; or each of the maleficent planets harming one of the luminaries; or one of them in opposition to one of the luminaries and the other rising to the place of the other by way of conjunction, being then, as we have described, in the parts that follow the parts of the luminaries: then the child has no life. For the amount of harm in this matter corrupts and hides the benefit and soundness in prolonging life that come from the distance of the place that rises to the place of the luminaries. What harms the sun in particular, of what is in the place that rises to its place, [is Mars], and the moon, Saturn; but in what faces them on the diameter, or overpowers them, the matter is the reverse: Saturn harms the sun and Mars the moon, especially if they have dominion over the place of the luminaries and the ascendant by governing them. If there are two oppositions on the diameter, and the luminaries and the two maleficent planets are on the cardines in an isosceles configuration, the infants are then born either dead or in a state between life and death.[75]

f. 44r

When these things are so, and the luminaries are separating from one of the beneficent planets, or share a configuration with it, and the rays reach the preceding parts, the child lives for the number of the signs between the haylāj and the nearest ray of the maleficent planet, as so many months, days or hours, according to the amount of the badness of the condition and the strength of the things that are its causes. If the rays of the maleficent planets reach the preceding parts, the

[72] Ptolemy has dogs and cats among the sacred animals; the Arabic has النموس, ichneumons. The manuscript leaves a space where the editor supplies ⟨human beings⟩ from the Arabic *Tetrabiblos*. The names Hermaphroditos and Harpocratiac are read from the Arabic *Tetrabiblos* (ارمامرويطس، ارقوقراطاقوى) for ارماطروديطس and ادفوفراطيافوا; Hermes and Aphrodite are Mercury and Venus. «An interpreter of dreams», «surround» and «of learning and of cunning» follow the Arabic *Tetrabiblos* and the Berlin manuscript (مفسّر الأحلام، المحفوفة، أدب، مكر); «their earnings» is written in the margin. Ptolemy speaks here of prophets who make money by it.

[73] *Tetrabiblos* III.9.

[74] «Kinds», «distinct from it», «shorter than» and «in potency» follow the Arabic *Tetrabiblos* (الأجناس، متبايناً له، دون، بالقوّة).

[75] The words from «the place of the luminaries in prolonging life» to «rises to its place» are written in the margin; the name of Mars is missing.

child lives for the number of the degrees between the haylāj and the nearest ray of the maleficent planet, as so many months, days or hours, according to the amount of the badness of the condition and the strength of the things that are its causes. If the rays of the maleficent planets reach the parts preceding the luminaries and the rays of the beneficent the parts that follow them, the child is nourished and lives. Again, if the maleficent planets prevail over the beneficent ones that share in the configuration, the child will be of a bad condition and abased; if the beneficent overpower, the child passes to others than its parents. If one of the beneficent planets rises with the moon or applies to it, while one of the maleficent is setting, the child is reared by its parent. In this way the consideration is made for multiple children: if one of the planets that share in configurations two by two or more is in the west, the child is born between dead and alive, or like a lump of flesh, or deficient and incomplete in form; and if one of the maleficent planets overpowers it, the child is not reared and has no life, as that requires.[76]

f. 44v

The ancients also said about the rearing: Look at the [lords of the] triplicities of the ascendant and of the sun by day, and at the lords of the triplicities of the moon and of the lot of good fortune by night. If you find the greatest of them, or the one of greatest authority among them, in good places of the sphere and of the signs — especially the middle of the sky, the eleventh and the fifth — clean of misfortunes and defects, it signifies rearing and life. If Venus is clean of defects and is in the fifteen degrees above the ascendant, and Jupiter, being clean, looks at it, it signifies rearing. If the moon is not afflicted, and none of the maleficent planets is in the cardines, and in the ascendant, the middle of the sky or the eleventh there is one of the diurnal planets by day or of the nocturnal by night, it signifies rearing. When the lords of the triplicities are in the middle of the sky, the eleventh and the fifth, they signify ease in rearing; when they are in the seventh, «والفنّ», difficulty (?) in rearing, especially when they are maleficent. When the cardines are upright, and the planets free from misfortunes, clean of defect, in soundness of their mixtures, and the lords of the houses in good places, in the terms of the beneficent planets, and their rays and the luminaries agree with one another and with the ascendant, it signifies long life. Look at the planet of greatest authority among the lords of the triplicities of the ascendant and of the sun by day, and of the moon and the lot of good fortune by night, whether it be one or two. If it happens to be one, and it is fallen and conjoined with the degree of a maleficent planet, and the degree of the ascendant is also conjoined with the moon, it is an indication of short survival. Likewise when a maleficent planet is conjoined with the degree of the ascendant, or with the degree of the luminary of the domain, and likewise with the lords of the triplicities of the luminary; and likewise when all the lords of the triplicities of the ascendant decline, and the luminaries and their lords decline, and the conjunction or the full moon and its lord decline: then there is no survival for the child. If the dominant planet has fallen from the cardines and is afflicted, but there are some degrees between it and the maleficent planet, the child survives for as many days, months or years as there are degrees between the dominant planet and the maleficent.[77]

f. 45r

Likewise when the maleficent planet is in the cardine and the dominant planet has fallen from the cardine, or the degree of the ascendant or the degree of the moon is looked at by a maleficent planet: the child survives for the amount of those degrees, as days, months or years. When the moon is afflicted and in a cardine, especially the ascendant and the middle of the sky, and no beneficent planet looks at it, it signifies short survival; especially if the maleficent planet is with the degree of the moon or the degree of the ascendant, or in quadrature or opposition to them, in the degree and the term; and especially if it is with the degree of the ascendant and of the

[76] «Separating» reads منصرفين for مضرّين, «harming»; «for the number», «the maleficent planet», «prevail» and «abased» follow the Arabic *Tetrabiblos* (مدّة، المنحس، غالية، ذليلًا). The second sentence repeats the first with «degrees» for «signs»; Ptolemy counts degrees.

[77] «Clean» reads نقيًّا for بقيًا; «a maleficent planet» النحس for الحسن, «the good»; «short» قلّة for فلة; and «no survival» لا إبقاء for الإبقاء, «survival».

moon together: for when it is so, even if the beneficent planets then look at it, the child does not live. If the moon is in the seventh, afflicted by bodily conjunction, quadrature or opposition with the degree of a maleficent planet, and no beneficent planet looks at it, judge with certainty that the child will not live. If the moon is in the fourth cardine and the maleficent planet faces it, especially if another maleficent planet is together with the moon, the child and its mother will not both escape death on the day of the birth; unless the moon is in its house, its exaltation or its triplicity, and a beneficent planet looks with strength at that maleficent one from the trine: that signifies that both escape, because a beneficent planet in the place of its dignities increases in strength and good, and a maleficent planet in the place of its dignities decreases in evil; especially when Mars is in a feminine sign and Saturn in a masculine sign. If the moon is as I have described, besieged between two maleficent planets, with no beneficent planet looking at it, and it is besides decreasing in light, then besides signifying the death of the child it signifies the loss of its sight before that. If there is a maleficent planet in the ascendant and one in the seventh, and the moon is in the cardine of the ascendant or of the middle of the sky, and the beneficent planets do not look at it, it signifies, with death, that the child is put out of its parents' house before it dies; especially if the moon is in the term of the maleficent planets and the lord of that term is in a cardine, for that is worse and more harmful. If most of [the lords of] the triplicities fall, especially in the sixth and the twelfth, and above all the one of greatest authority among them, they signify short survival; and the time in that is [found] by looking at the beneficent planet, when it is in a cardine or in a good, strong place, or at the planet that applies to the degree of the ascendant, of the luminaries and of the lot of good fortune: it signifies [the life] by the amount of its period (?), in years, months or days. When the maleficent planets are in the ascendant, near it in degrees — even if there are beneficent planets there nearer in degrees to the degree of the ascendant — if the other significators are strong for rearing and life, it signifies that the child will be deficient in mind and limbs, with many injuries to the end of its life, and in any case will only be of short life. When you find the sign in which the conjunction or the full moon is, the lord of the term that looks, which is called «المعنق», the lord of the term of the luminaries and of the ascendant, and the lords of the triplicities that we have mentioned — especially the triplicity of the sun by day and of the moon by night — and the lot of good fortune, all afflicted, judge that the child will come out of its mother's womb dead, or come out dead from its mother's womb.[78]

## ⟨24⟩ On the length of life[79]

f. 45v

Of the accounts of the accidents that come after birth, the first is the account of the length of life. The consideration of it is not absolute and unrestricted, but is taken and known from the dominion and prevalence of the chief places, and it is something in which many ways are used. The way that agrees with our opinion and follows the natural course is the one that I shall mention, and I say about it: the consideration is from the places of the haylāj, from the places that have dominion over the places of the haylāj, and from the killing places or planets too; and each of these matters is distinguished as I shall describe to you. The first thing we should know is that the places of the haylāj are those in which the planet to which the authority of the haylāj falls must necessarily be. They are the sign of the ascendant, from the 5 degrees that precede it and rise from the horizon before it to the remaining 25 degrees that rise after it and ascend; the [thirty degrees] to the right of these thirty degrees, in sextile to them, which belong to the house of good fortune; those in quadrature to them, which belong to the middle of the sky; those in trine to them, which belong to the place called Theos (?), which is the house of religion (?) and

[78] «Is in» reads وهو for ونحن, «and we»; «bodily conjunction» مجاسدة for محاسدة; «dignities» حظوظه and الحظوظ for خطوطه and الخطوط, «lines»; «a maleficent planet in the place of its dignities» النحس for الشمس, «the sun»; «a feminine sign» برج for موج, «wave»; «the luminaries» النيّرين for المضيين.

[79] *Tetrabiblos* III.10. The *haylāj* is Ptolemy's prorogator, the significator of life (see the glossary of the *Zīj*); the *kadkhudāh*, below, is the planet that gives the years.

the fortunate; and those opposite them, which belong to the setting point.[80]

Of these, what should be put first for its strength in dominion is what is in the middle of the sky; then what is in the ascendant; then what is in the house of good fortune, which is the one that rises to the middle of the sky; then what is in the setting point; then what precedes the middle of the sky, which is the ninth, of which we said that it is the house [of religion]. All that is below the earth we should leave unused in so noble a matter, except the 25 degrees that are always rising and appearing above the earth. Of what is above the earth one should [not] use the sign that has no connection with the ascendant, nor the sign that has risen before the ascendant, which is called the house of misery: for besides its declining from the cardine, the power that goes out from it to the earth is troubled by the thick, dark vapour that rises to it from the moistures of the earth before the rising of the sun; and because of that the colours of the stars in this house, and the sizes of their bodies, are seen otherwise than they are by nature.[81]

f. 46r

Then we should take as the haylājs the four chief things, I mean the sun, the moon, the ascendant and the lot of fortune, which is the lot of good fortune, and the governors of the places of these things. The lot of fortune is found by always taking the number [of degrees] from the place of the sun to the place of the moon, both by day and by night, and beginning from the ascendant and counting from it the same number, proceeding in the direction of the following signs: so we know the place of this lot. We do that so that the relation of the sun to the ascendant may be like the relation of the moon to the lot of fortune, and the lot becomes as it were the ascendant of the moon. Likely it was this very meaning that was intended by the one who said that in nativities that occur at night one should count from the moon to the sun [what one counts] by day, and then begin from the ascendant and count from it the same number in the direction opposite to the one we counted by day, I mean towards the preceding signs. For when we do that, the place of the lot of fortune comes out for us in this way as the very place that we had, and the condition in the sharing of the configuration becomes the same condition.

f. 46v

Of these things one should choose, by day, first the sun, if it is in the places of the haylāj; if it is not so, the moon is chosen; otherwise the planet that has the most shares in the governing of the place of the sun, of the place of the preceding conjunction and of the ascendant — I mean when it has, of the five shares on which the governing rests, three shares in one of the places we have mentioned, or more than three, or has them in all of them; and if that cannot be, we take the degree of the ascendant. By night one should choose first the moon; then the sun, if it is in the signs that rise and ascend above the earth from the house of the ascendant; then the planet that has the most shares in the place of the moon, the place of the preceding opposition and the place of the lot of fortune. If that cannot be, we take in the last resort the ascendant, if what preceded the birth was a conjunction, and the lot of fortune, if what preceded was an opposition. If both the luminaries, and the governor according to the suitable domain, are in some of the places of the haylāj, we should take whichever of the luminaries is in the place of greater authority and strength, which is clear from the order we have set out for the places of the haylāj. The governor should be chosen over the luminaries only when it is in a place of greater authority and strength among the places [of the haylāj], in that order, and has shares in the governing according to both domains. By this procedure the haylāj that signifies the length of life is known.[82]

Others than Ptolemy among the ancients agreed that none of these that we have mentioned deserves to be haylāj unless the lords of the shares in its place witness to it. The witnessing is that they share in a configuration with it by one of the sharings, whichever it may be: quadrature,

[80] «Killing» reads القاتلة for المقابلة, «opposite»: Ptolemy has the destructive places or stars. Ptolemy calls the eleventh place the House of the Good Daemon and the ninth the House of the God (θεός); for the transcription's باوس the Arabic *Tetrabiblos* has ثاوس, and for بيت الدين والمسعد the Berlin manuscript has بيت الذين والسعر.

[81] «The middle of the sky» is written in the margin. Ptolemy calls the twelfth place the House of the Evil Daemon.

[82] «In the last resort» is written between the lines over سائر; «of greater authority» and «the luminaries» follow the Berlin manuscript (أكثر، النيّر).

opposition, trine and sextile; or looking, or equality in power; or of the commanding and obeying signs, which are called the high and the low. The one of them that has this witnessing is the haylāj, and the one among the lords of the shares in its place that shares [a configuration] with it is the giver (?) of the years. We have seen that Ptolemy mentions in these places the lords of the shares, which are those in charge of the governing of these places, but does not afterwards call them to witness for the haylāj. It may be that Ptolemy hinted at it, but that it did not come out in the translation, or the like. Those who held this view put the degree of the conjunction or of the full moon that precedes the birth, in the haylāj-ship, in the place of the planet that has the shares in those places, and made no distinction between above the earth and below it in the places of the haylāj. We shall explain their opinion when we have finished with Ptolemy's opinion in this chapter.[83]

f. 47r
f. 47v

Ptolemy said: Once we know the haylāj, we should use the two kinds of direction, the kind that goes in the order of the signs, and the one that goes both in the order and against the order. The order of the signs alone should be used in the condition called the casting of rays, that is, when the haylāj is in the eastern places, I mean those from the middle of the sky to the ascendant. The one used both in the order of the signs and towards the preceding signs together is in the condition called the «hourly», which is when the haylāj is in the ninth place, called Theos, declining from the middle of the sky. This being so, the killing [degree], according to the haylāj [directed] by the kind towards the preceding signs, is only the degree of the setting point, because it hides the lord of life. The degrees of the planets that meet the haylāj or witness to it only decrease and increase the numbers that accumulate up to the time of the setting of the haylāj, without killing; for it is not they that come to the place of the haylāj, but it comes to their places. Those that increase are the beneficent planets and those that decrease the maleficent. Mercury goes with those of these planets that share in a configuration with it: if they increase, it increases, and if they decrease, it decreases. The number of the increase and the decrease is known from the position of the degree of each of them, or of its ray: for the number of the full years is according to the times of the hours of each degree of them, by the times of the day hours if the time is diurnal and by the times of the night hours if it is at night. This must be understood as said of them in the ascendant; [the number] then decreases according to its distance from the ascendant, until it reaches the setting point and nothing of it remains. This is found by taking the distance of the degree from the setting point, as we have explained in the first chapter of this book on the hours of the distance of the degrees from the cardines: for each seasonal hour of it take a twelfth of the times of the hours of that degree, nocturnal or diurnal according to the time, and what comes out is the amount of the increase or the decrease.[84]

f. 48r

According to the kind that goes towards the following side from the place of the haylāj, what kills is the places of the maleficent planets, I mean the place of Saturn and of Mars: either when they meet the haylāj with their bodies, or when they cast their rays from either of these two places, I mean the place of quadrature and the place of opposition; and sometimes this is from the sextiles that hear and obey, or that look and are equal in power. The very quadrature of the place of the haylāj that is on the following side of the signs kills too; and sometimes the sextile does so, when it is afflicted and in signs of long ascensions, and the trine does it when it is afflicted and in signs of short ascensions. When the moon is the haylāj, the place of the sun may kill too. One should not suppose that these places necessarily kill at every time; they do so only when they are afflicted. For it is prevented too if the term belongs to one of the beneficent planets, or if one of the beneficent planets casts its ray, from quadrature, trine or opposition, either upon the killing degree itself or upon the degrees that follow it, provided that it does not exceed 12 degrees in the casting of the ray of Jupiter and 8 degrees in the casting of the ray of Venus; and likewise when

[83] «Giver» guesses at the transcription's العين. Al-Battānī reckons with the Arabic translation of the *Tetrabiblos* used for this commentary, and allows that it may have lost something of Ptolemy's meaning.

[84] «Direction» follows the Berlin manuscript (التسير) for {التكيسير}; Ptolemy's «hourly» is the *horimaea*. Al-Battānī reckons the distance of a degree from the cardines in seasonal hours in chapter 55 of the *Zīj*.

the haylāj and the planet that meets it meet bodily, and the latitude of both is not one and the same. When the planets that release and help, or those that kill, are two planets or more on each of these sides, one should look at which of the two kinds prevails, by the number of those that help it and by their strength: by number, when one of the two kinds is more numerous than the other by something perceptible; by strength, when the planets that help or those that kill are in the places suited to them and the others are not in such places, and especially when the planets of one kind are eastern and those of the other kind western. In general, none of them that is under the rays should be used, either for killing or for helping and releasing; except that, when the moon is the haylāj, the place of the sun itself is the killer, and it is used with the maleficent planet that is with it, when it corrupts and none of the beneficent planets releases it; and when a beneficent planet is with the sun, in the place of that maleficent one, as we said, it releases it.[85]

f. 48v

The number of the years that come from the distances between the place of the haylāj and the killing place should not be taken absolutely, nor however it happens, as many people took it, who claimed that it always comes from the times of the ascensions of the degrees of that distance in the clime; and we should not use that at any time, unless the ascendant itself is the place of the haylāj, or one of the places that come to the ascendant, when the degree of the ascendant is the haylāj. When it is in the middle of the sky, I mean the degree of the middle of the sky, we use for the amounts of the distances the ascensions of the right sphere. When the place of the haylāj inclines from the cardines to one of the sides, we use what we described in our first book, where we mentioned the ascensions of the signs between the quarters of the sphere, so as to know after how many times of the equator the following killing place comes to the place of the haylāj, which is the preceding place; and when it reaches the setting point, the amounts of the distances are according to the settings of the degree of the setting point, when the haylāj is in it. That is something we have explained, and we can do without mentioning it here, since we finished with it in the book, because it is one of the geometrical things whose working had to be explained there.[86]

f. 49r

It remains for us to distinguish each of the kinds of things that meet the haylāj and its setting: of the things we have mentioned before, those that are shortest in time, those that kill, those that are called climacterics, which are the [critical] times, and the other things that meet it. Distinguishing this comes from knowing the bad, harmful meeting and the good, beneficial, helping one, which occur ⟨in⟩ the way we mentioned just now; and from the ingress of the coming years, which are the times of the revolutions afterwards that the place that the haylāj meets signifies. For when the places of meeting are bad and afflicted, and the positions of the planets at the ingress of the following years harm the chief places, we should suppose that what happens then is death in truth; when one of these two matters is sound, we should suppose that what happens is injuries of great danger; and when both these matters are sound, we should suppose that what happens is sluggishness, weakness, harms and exile, by departure or travel. Instead of the word «exile» one may say «decline of rank, life abroad and journeys». The particular kinds that arise from these things are known from the places of the things that meet [the haylāj], for the matters of the native. Nothing prevents us, when we are in doubt at some times about the places that must have the matter of killing, from reflecting on and considering the things that meet the haylāj in each of the matters of the native: one should [look] in what comes in them for what agrees with the accidents that have already occurred and appeared, and examine the matter in all of them, when the matter, in the completion of what comes from them, is equal in strength in all of them, and so make our examination of their increase and decrease accordingly. One should correct the amounts of the applications and the distances between the haylājs and the planets according to the latitudes of the planets, as we have explained in the chapter on establishing the

[85] «Killing degree» reads القاتلة for المقابلة; «not one and the same» follows the Arabic *Tetrabiblos* (لم يكن) for وليكن; «kill» reads تقتل for يقبل.

[86] «Our first book» is al-Battānī's *Zīj*, chapter 55, «On knowing the ascensions of the signs between the cardines, in the quarters of the sphere». The words «from the times of the ascensions» stand twice.

applications and the distances between the planets, before directing from the haylājs.[87]

f. 49v
f. 50r

Those who followed the doctrine we have described said about the giver who gives the years, the kadkhudāh — I mean the lord of the house, the exaltation, the triplicity [and the term] of the haylāj, putting the lord of the term first — that whichever of them looks at the haylāj is the kadkhudāh; or if two look at it together, the nearer of them in degrees, before or after, is the kadkhudāh. This is when each of them has a share in the ascendant, the lot of good fortune and the sign of the conjunction or the full moon; then it goes to the one in charge (?), whether its look is near or far, after you have looked at the degrees. If their testimonies are equal, the strongest of them in position; if they are equal in position, the nearest of them to the cardine or to what follows the cardine. If some of them are foreign and some in their dignities, the one in its dignities has the better claim, and the eastern, the farther from the sun, comes first. Know that when the sun is in Aries or Leo, it takes over both the haylāj-ship and the kadkhudāh-ship, and likewise the moon in Taurus or Cancer, because each of them has two dignities in these two signs — when there is no kadkhudāh looking at it. When you know the kadkhudāh: if it is in the ascendant or in the middle of the sky, eastern, in its dignities or in its suitability of masculine and feminine and of night and day, free from retrogradation and burning, [it signifies its greater years]; if it is in the eleventh or the fifth, exceeding what I have mentioned by one quality, it signifies its middle years; if it is in the second or the eighth, as I have described for the ascendant and the middle of the sky, or in the sixth or the twelfth, as I have described for the eleventh and the fifth, it signifies its lesser years. Whenever a quality is lacking, the rank of what the kadkhudāh signifies, from whatever place it may be, comes down one step. For example, what I have described of its being in the ascendant or the middle of the sky: if it is not eastern, it comes down one step, and its greater years go back to the middle ones; if besides it is not in some of its dignities or its suitability, it goes back to its lesser years; if besides it is retrograde, it goes back to its lesser years as months; and if besides it is burnt, to its lesser years as days. Speak of all these places by this analogy. They said that when the Head is with the kadkhudāh within 12 degrees, before or after, and the kadkhudāh is beneficent, it adds a quarter of the years of the kadkhudāh that it gives; if the Tail is in the place of the Head, it takes away as much; and if the kadkhudāh is maleficent, the Head takes away from its years and the Tail adds to them by the same analogy. The nearer the degrees, the more certain it is in both cases. If the kadkhudāh is in the condition that signifies nothing, and Jupiter and Venus, or one of them, are in the ascendant or the middle of the sky, the native may hope for as many years as the lesser years of whichever of them happens to be in the cardine we have mentioned, provided that it is not burnt (?) by the maleficent planets, and the moon afflicted (?).[88]

f. 50v

When you know the years of the kadkhudāh, look at the planet that looks at the kadkhudāh. If a beneficent planet looks at it, free from the maleficent, from retrogradation and from burning, it adds as much as its lesser years. If a maleficent planet looks [at it] from opposition or quadrature, it takes away as much as its lesser years; if the maleficent planet has shares in the nativity and is eastern in some of its dignities, it takes away as many months as its lesser years; and if besides its look is one of trine or sextile, it takes away days only. Mercury, when it is with the beneficent planets and the givers, eastern or western, gives as much as its lesser years; and when with the maleficent and the diminishers, it takes away as much. The sun, in opposition and conjunction, takes away like the maleficent planets, and in quadrature it adds nothing but a little. The sun is

[87] «Climacterics» reads the Greek κλιμακτήρ from the Arabic *Tetrabiblos* (قليماقطر) for {طيمًا} قطير فاس; «kill» follows the Berlin manuscript (قاتلًا). The sentence «when one of these two matters is sound … death in truth» stands twice, the first time with the wrong conclusion. The chapter on applications is chapter 54 of the *Zīj*, «On establishing the amounts of the applications as they depend on the latitudes of the stars».

[88] «The giver» reads المعطي for المعنى, and «whichever of them» أيّهم for أي هو. «[It signifies its greater years]» supplies what the example that follows requires. The words «if besides it is retrograde, it goes back to its lesser years as months» stand twice. «Retrogradation» is written in the margin.

most cutting and most maleficent for the kadkhudāh when it is with Mars, because their heats come together. When the kadkhudāh is under the rays, it cannot receive what the beneficent planets add, unless it is in the heart [of the sun], for then it is at its most fortunate, whether it is maleficent or beneficent. When you know the amount of the indication of the kadkhudāh, direct the haylāj by the degrees of ascension for its place in the sphere, in that way, to the bodies of the maleficent planets and their rays, to the Tail, to the quadrature, the opposition and the conjunction of the moon, and to the degree of the setting point, if it has passed (?) what is below the earth: a year for each degree. If those years agree with the years of the kadkhudāh, or are near them, and no beneficent planet casts its ray to that term in which the haylāj reached the ray of the maleficent planet, the native perishes in that year; especially when the lord of the native's year is afflicted, or the lord of the ascendant afflicted by that maleficent planet. If it reaches another maleficent planet that does not agree with the years of the kadkhudāh, there is a calamity; and whenever the haylāj reaches the body or the ray of a maleficent planet, whether a beneficent planet casts its ray to that degree or not, it is a year of calamity; and that is worse when the maleficent planet is in the course of the degrees of the haylāj, that is, in latitude. The moon, when it is haylāj and applies to the afflicted ascendant or to the afflicted sun, kills; and likewise the degree …[89]

[89] «In the heart» renders صميمًا, the position within the sun's disc. «The amount» reads القدر for القمر, «the moon». «Kills» follows the Berlin manuscript (قتل). The text breaks off here; the manuscript, like the Berlin manuscript, leaves a large blank space before the next chapter.